

Dong-A Univ. (ISPL)



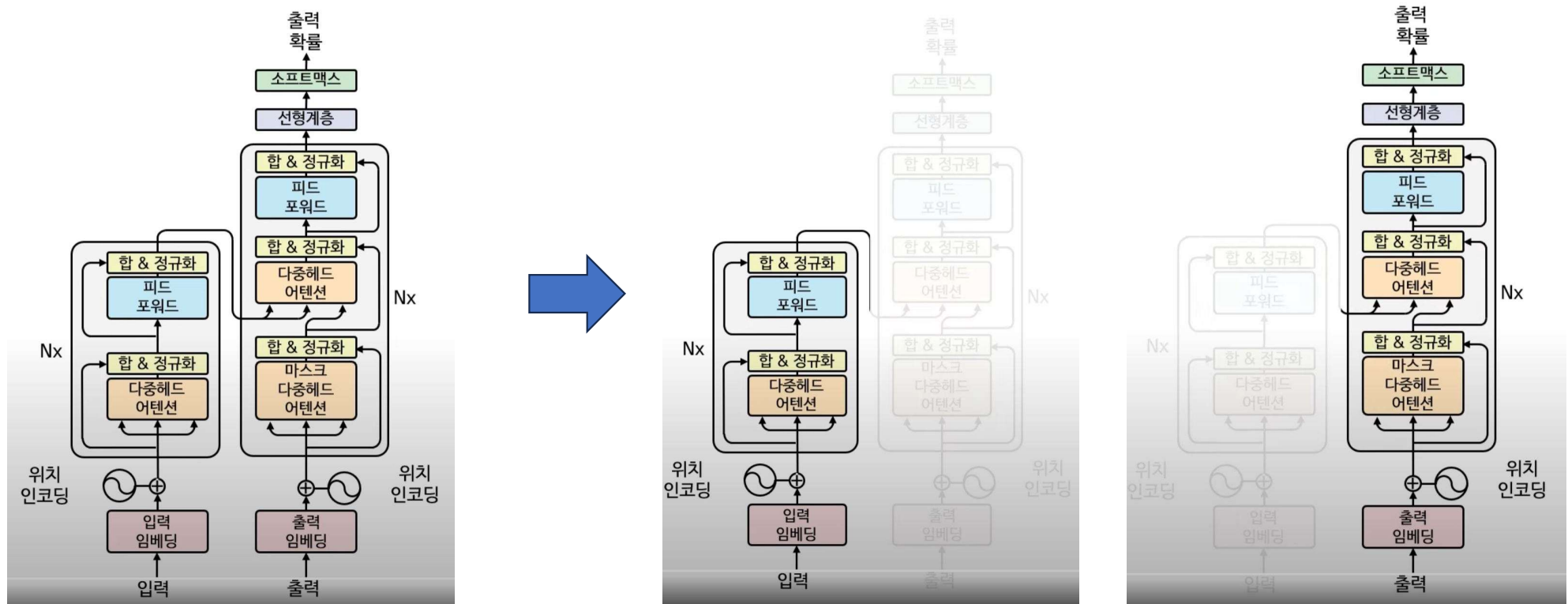
동아대학교
DONG-A UNIVERSITY

Transformer 예제

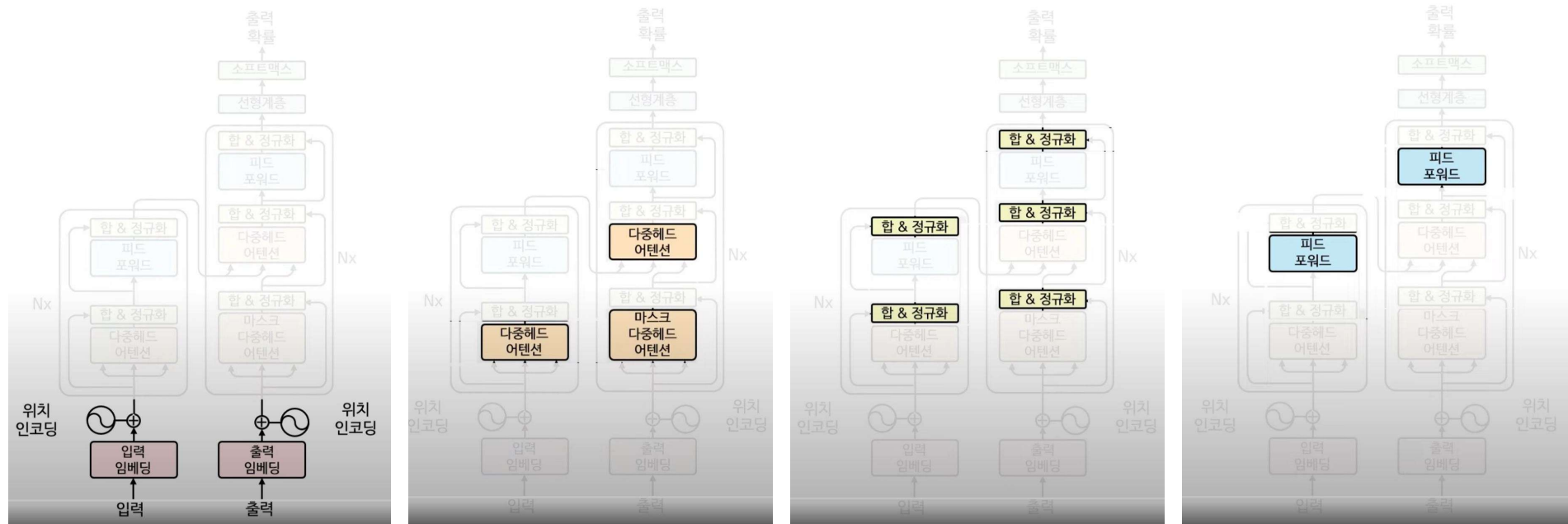
소프트웨어대학 AI학과
전 동 산



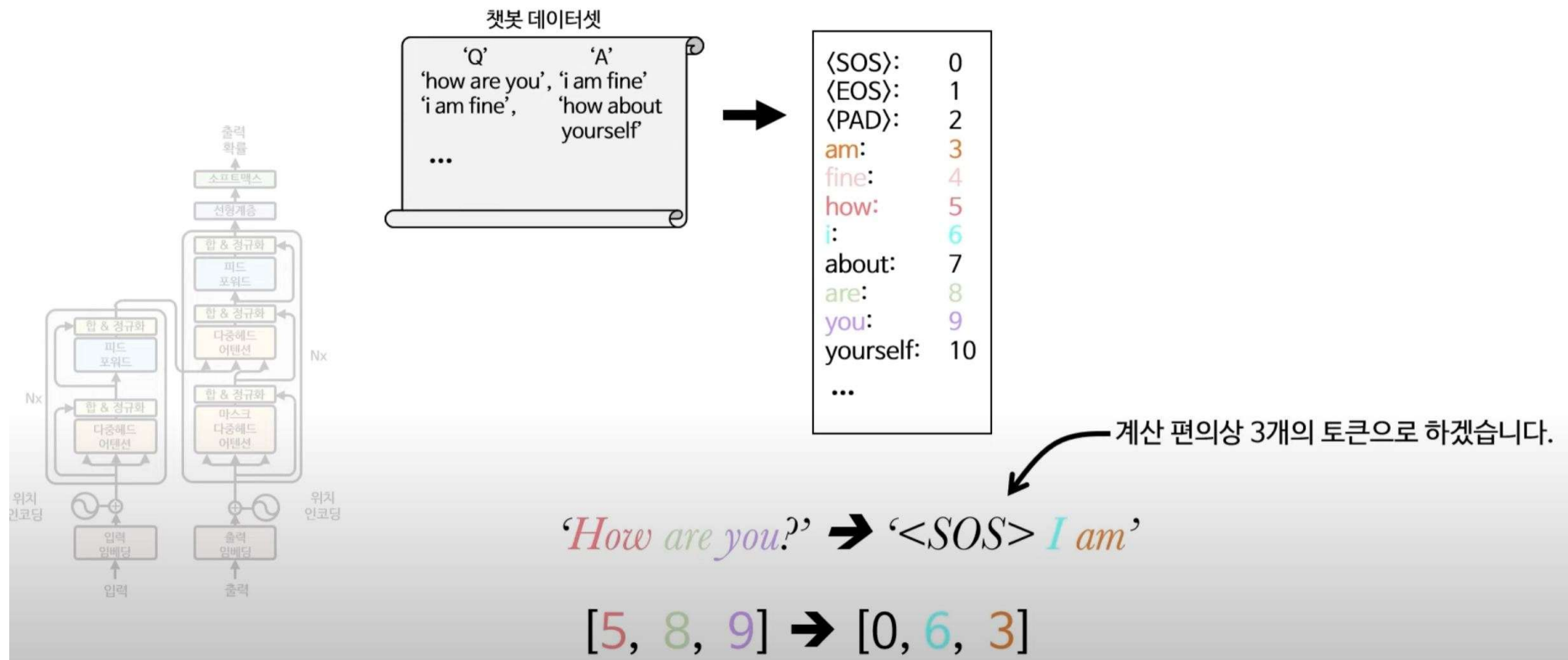
Transformer



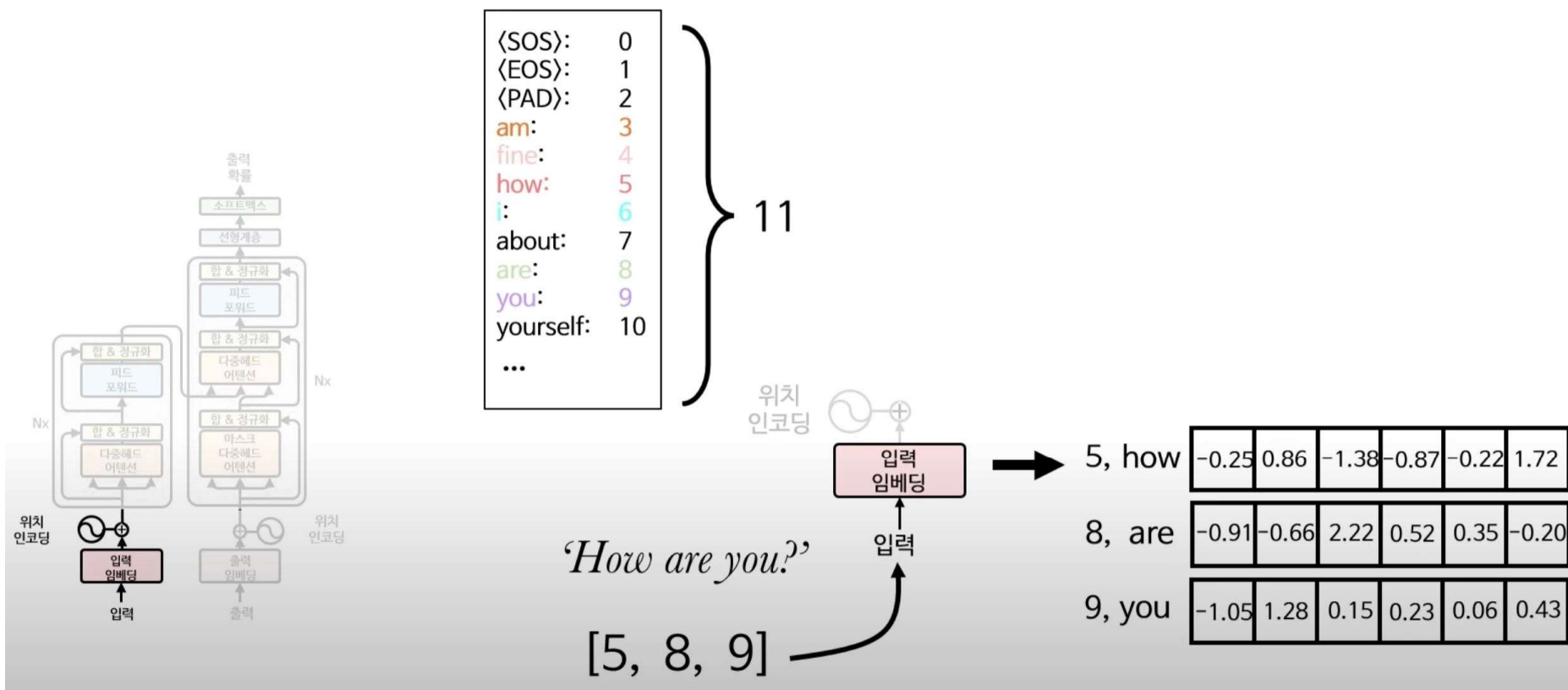
Transformer



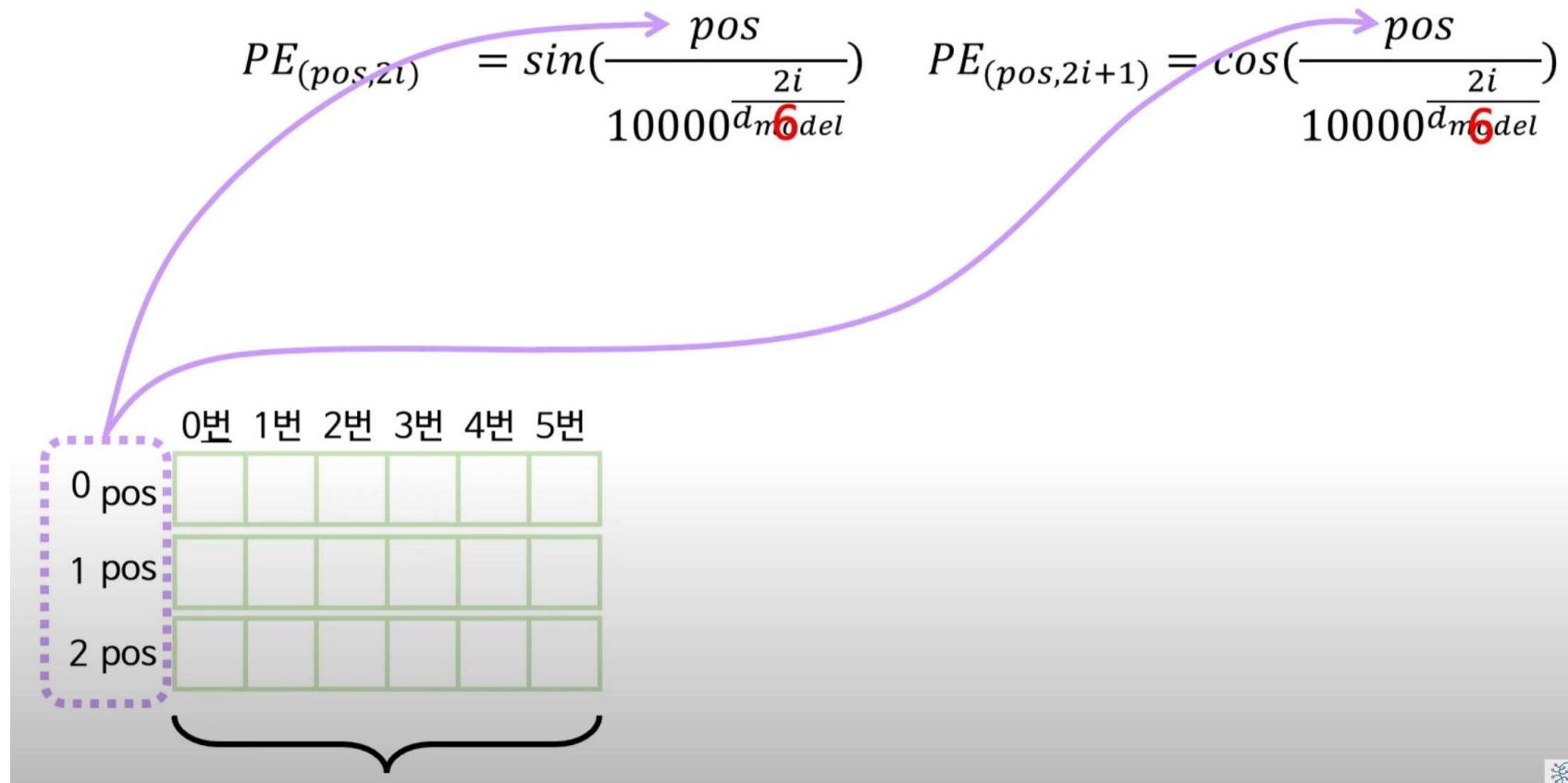
Transformer



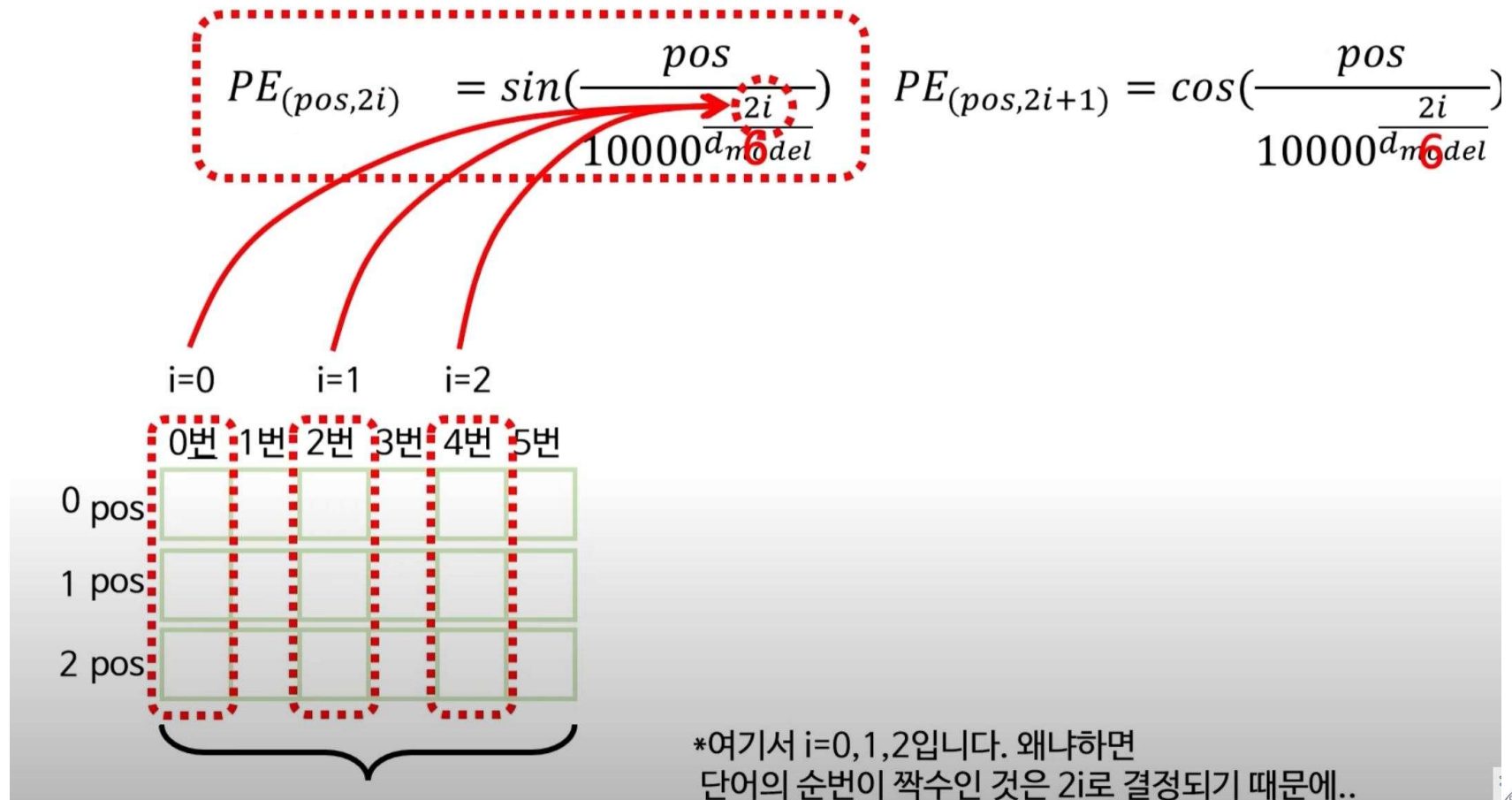
Transformer



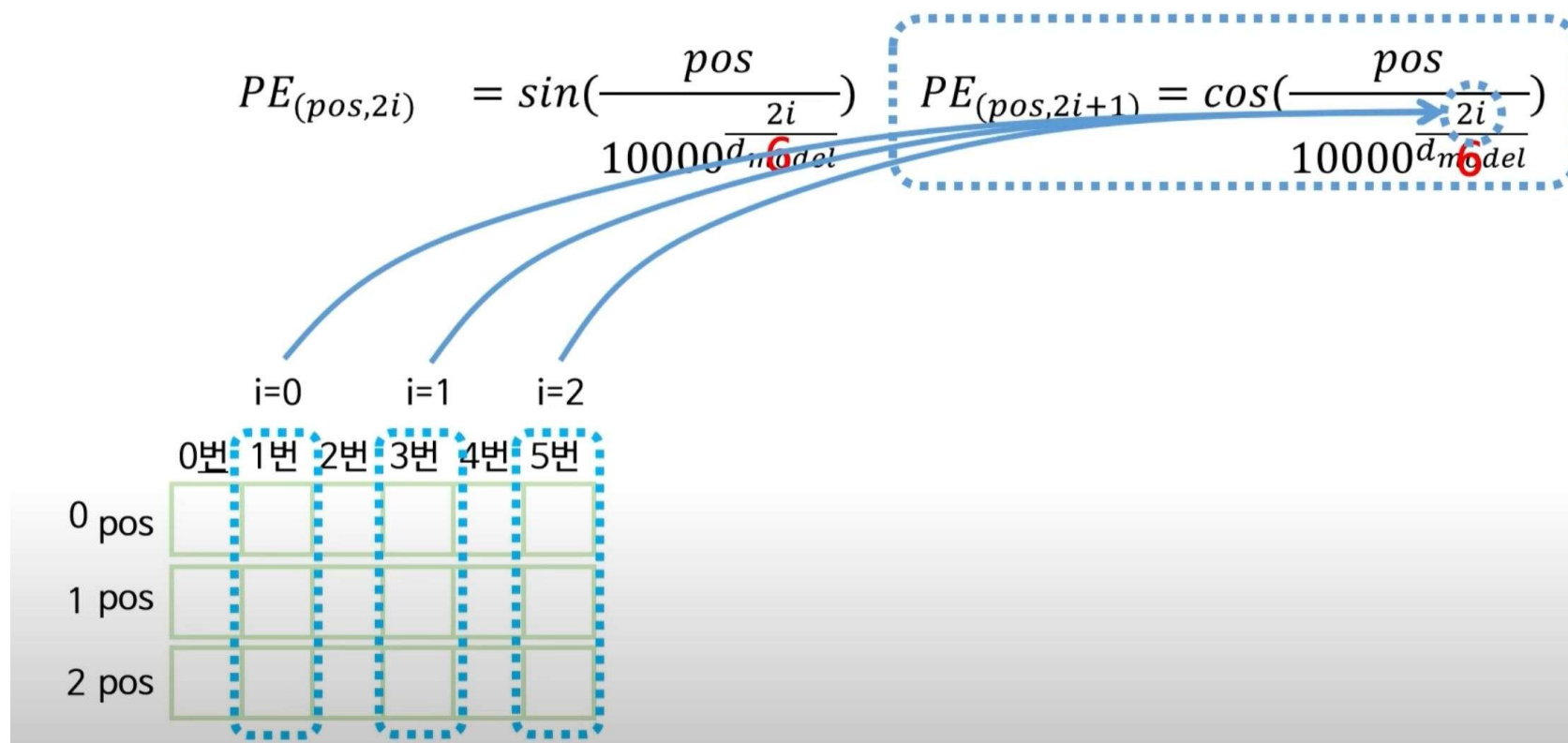
Transformer



Transformer



Transformer



Transformer

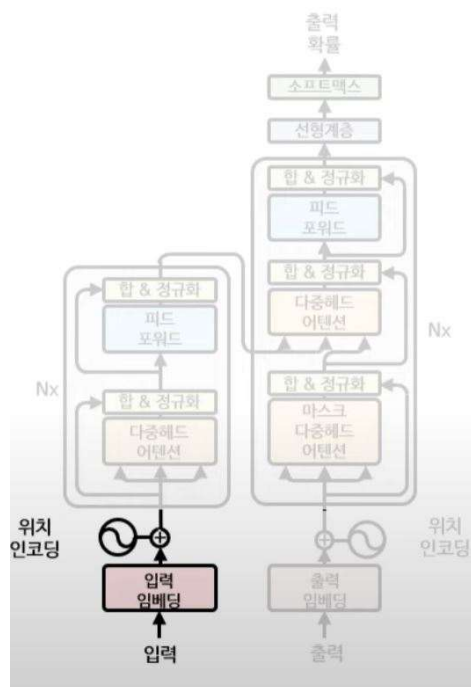
$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

	0번	1번	2번	3번	4번	5번
0 pos	0	1	0	1	0	1
1 pos	0.84	0.54	0.05	1	0	1
2 pos	0.91	-0.42	0.09	1	0	1

Transformer



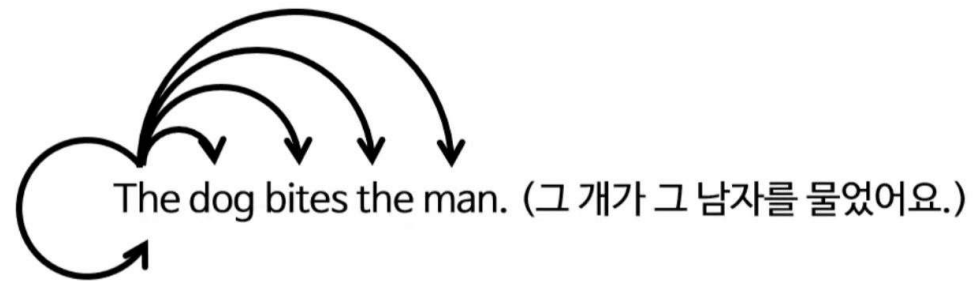
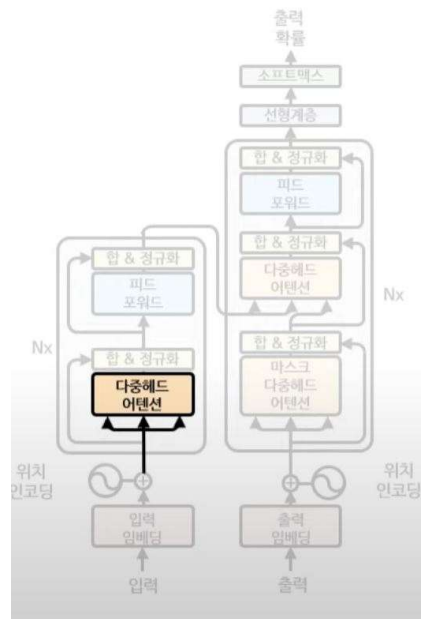
Transformer



-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

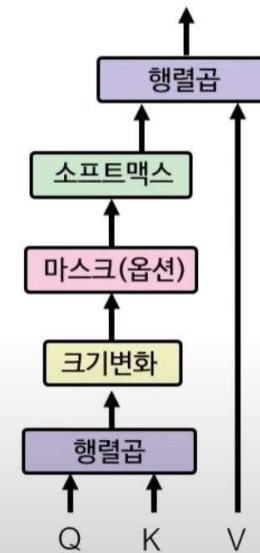
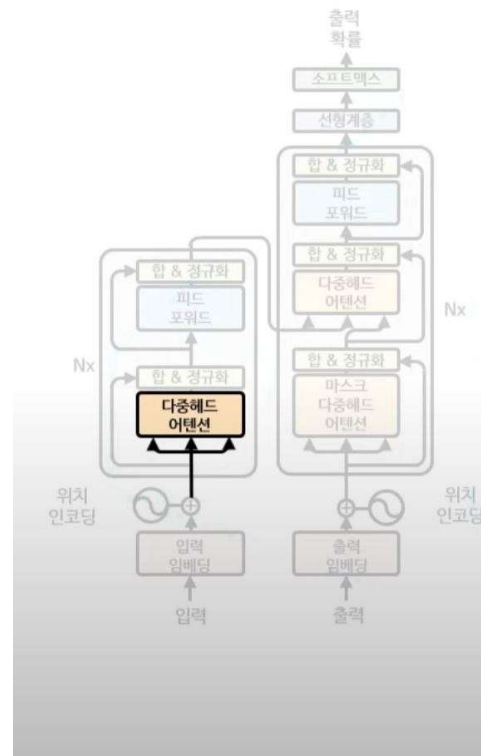
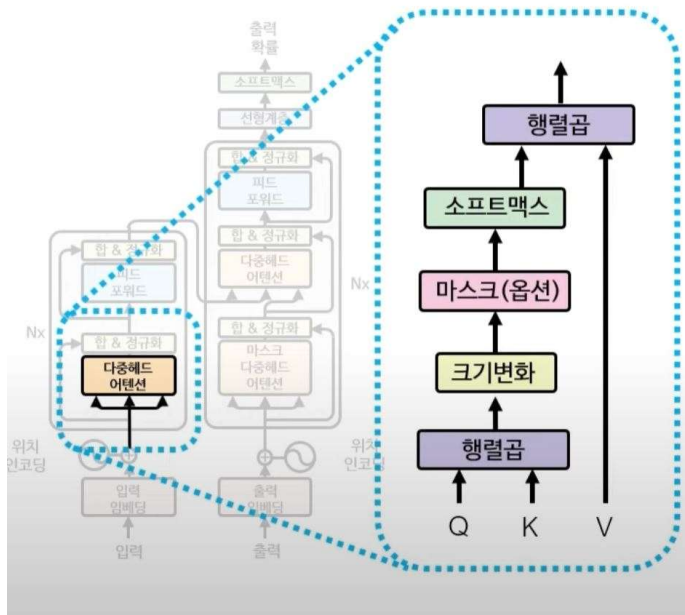


Transformer



Self-Attention

Transformer

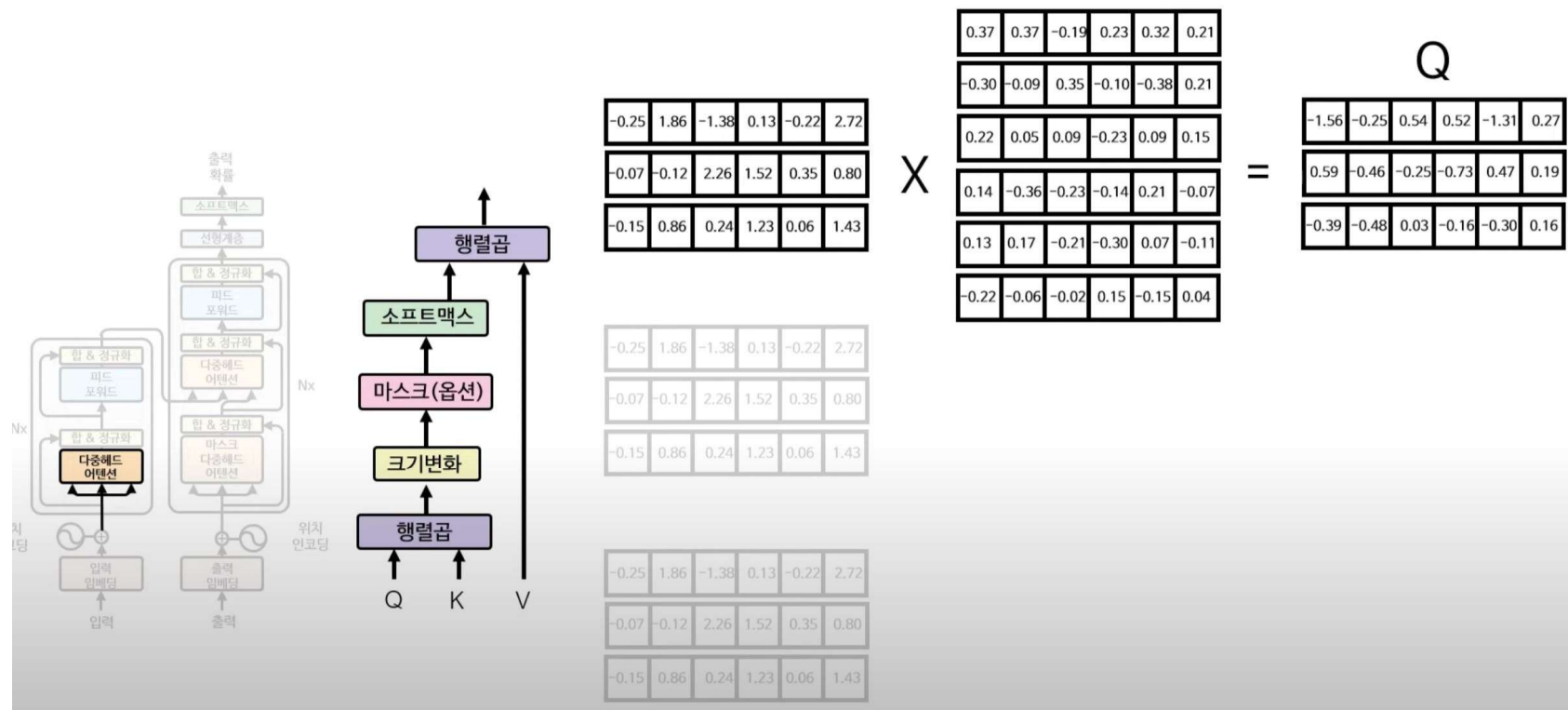


-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

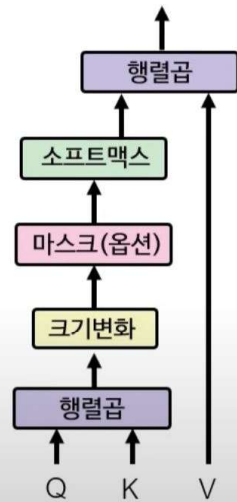
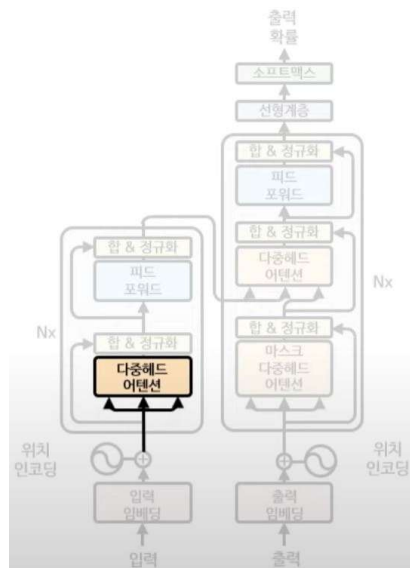
-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

Transformer



Transformer



-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

X

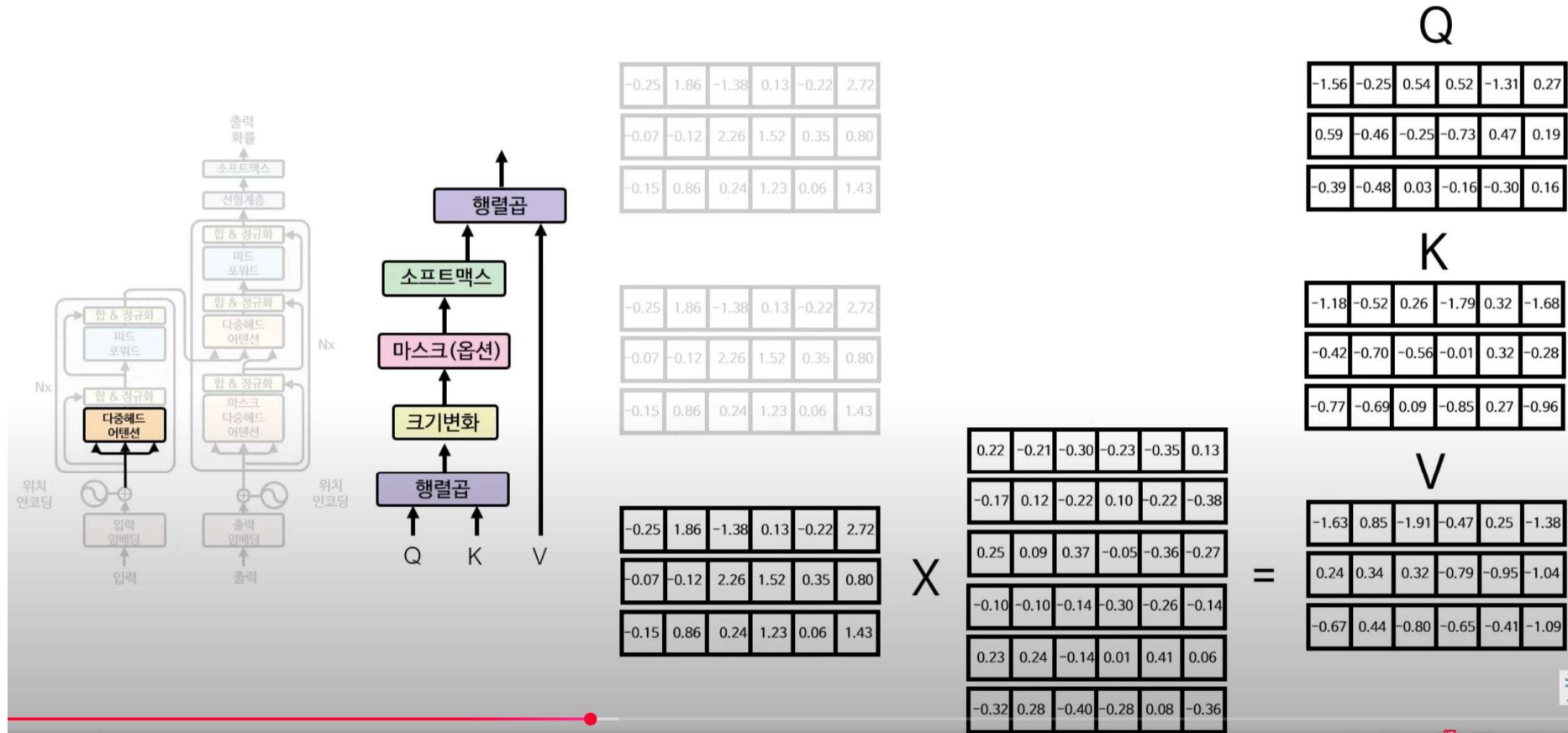
-0.18	-0.28	0.35	-0.34	0.30	0.37
-0.24	-0.24	0.13	-0.41	-0.30	-0.32
0.00	-0.14	-0.35	0.12	-0.07	0.10
-0.15	-0.32	0.28	-0.09	0.09	-0.18
-0.03	0.34	-0.11	0.16	0.21	-0.04
-0.28	-0.08	-0.16	-0.34	0.33	-0.31

=

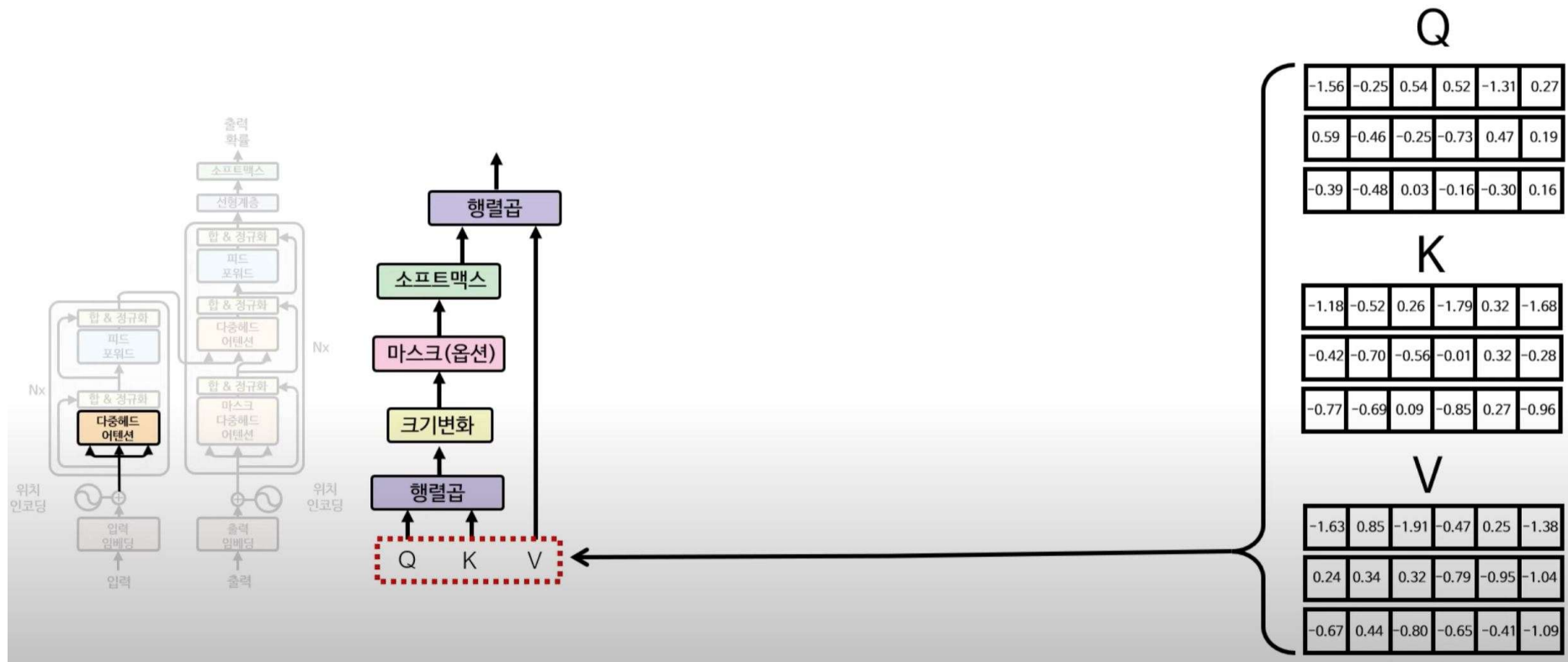
Q					
-1.56	-0.25	0.54	0.52	-1.31	0.27
0.59	-0.46	-0.25	-0.73	0.47	0.19
-0.39	-0.48	0.03	-0.16	-0.30	0.16

K					
-1.18	-0.52	0.26	-1.79	0.32	-1.68
-0.42	-0.70	-0.56	-0.01	0.32	-0.28
-0.77	-0.69	0.09	-0.85	0.27	-0.96

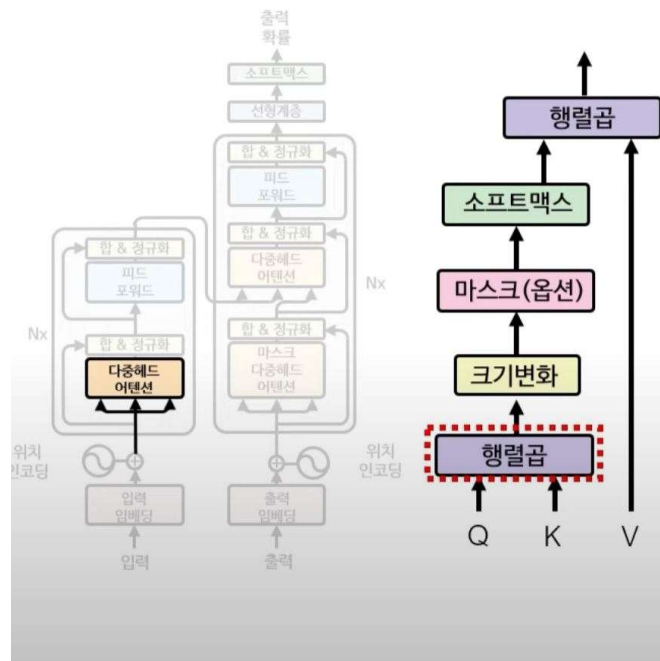
Transformer



Transformer



Transformer



$$Q \cdot K^T$$

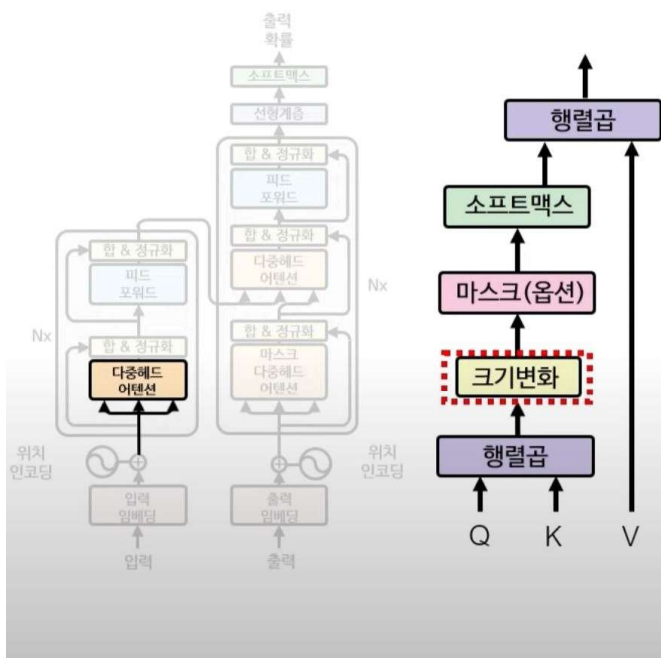
K (transposed)

Q						X	K (transposed)		
-1.56	-0.25	0.54	0.52	-1.31	0.27		-1.18	-0.42	-0.77
0.59	-0.46	-0.25	-0.73	0.47	0.19		-0.52	-0.70	-0.69
-0.39	-0.48	0.03	-0.16	-0.30	0.16		0.26	-0.56	0.09
							-1.79	-0.01	-0.85
							0.32	0.32	0.27
						-1.68	-0.28	-0.96	

V

-1.63	0.85	-1.91	-0.47	0.25	-1.38
0.24	0.34	0.32	-0.79	-0.95	-1.04
-0.67	0.44	-0.80	-0.65	-0.41	-1.09

Transformer



0.31	0.03	0.37
0.61	0.32	0.40
0.64	0.35	0.54

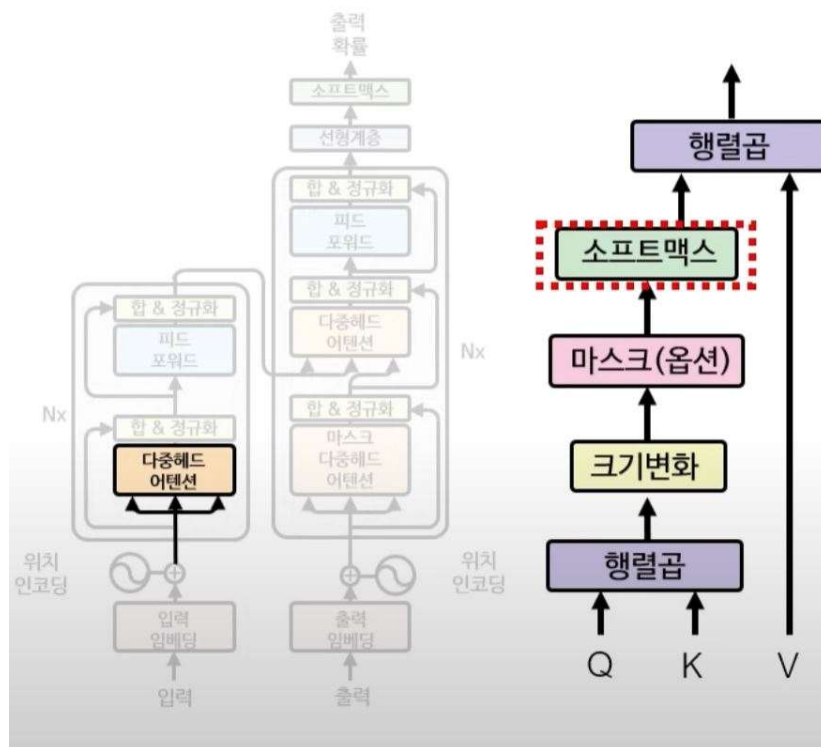
$\div \sqrt{6}$

이 예제에서는 d_{model} 의 길이가 6이기 때문입니다..

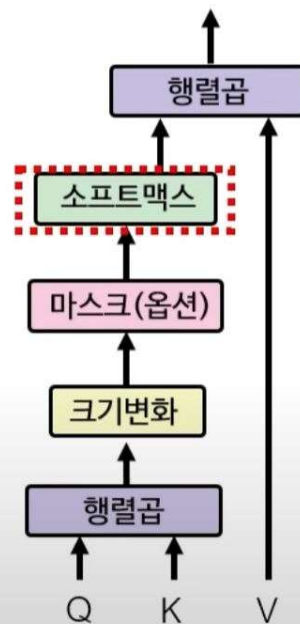
V

-1.63	0.85	-1.91	-0.47	0.25	-1.38
0.24	0.34	0.32	-0.79	-0.95	-1.04
-0.67	0.44	-0.80	-0.65	-0.41	-1.09

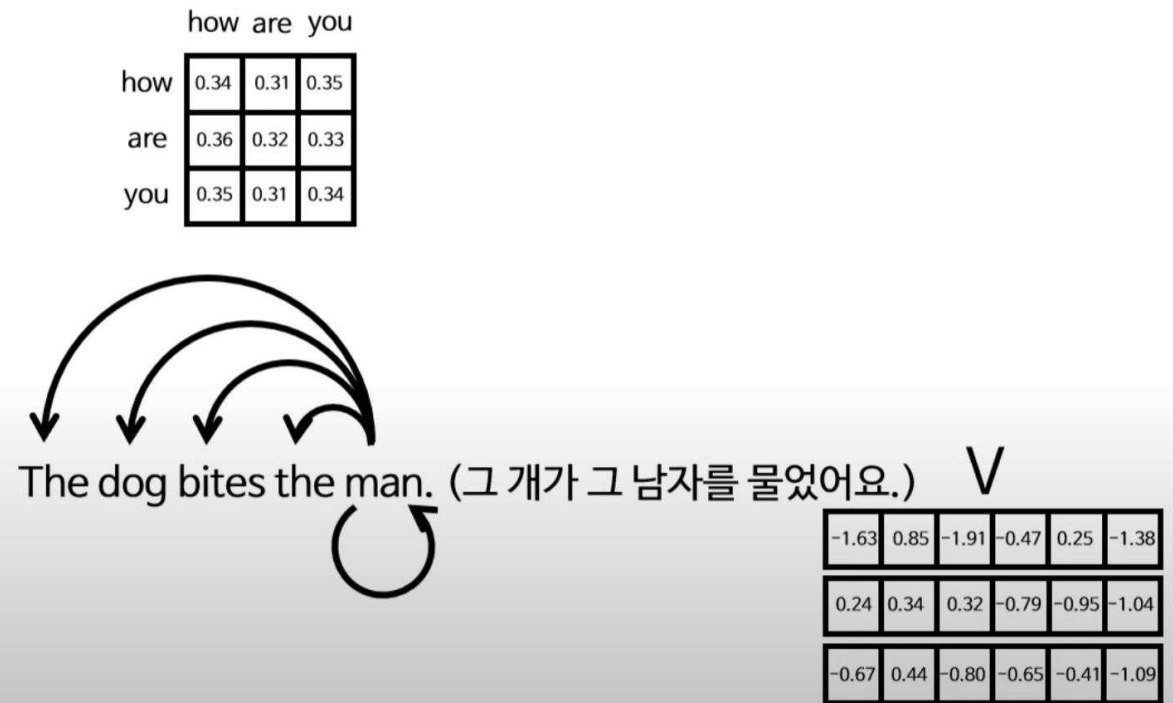
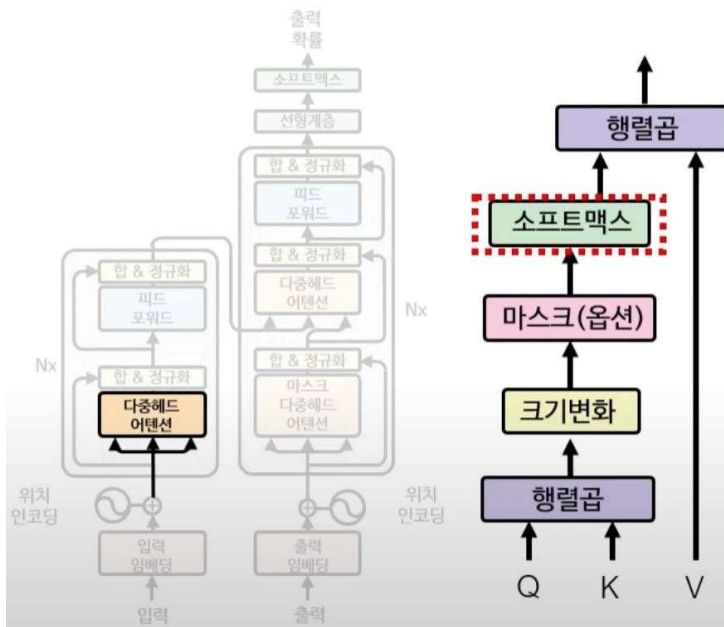
Transformer



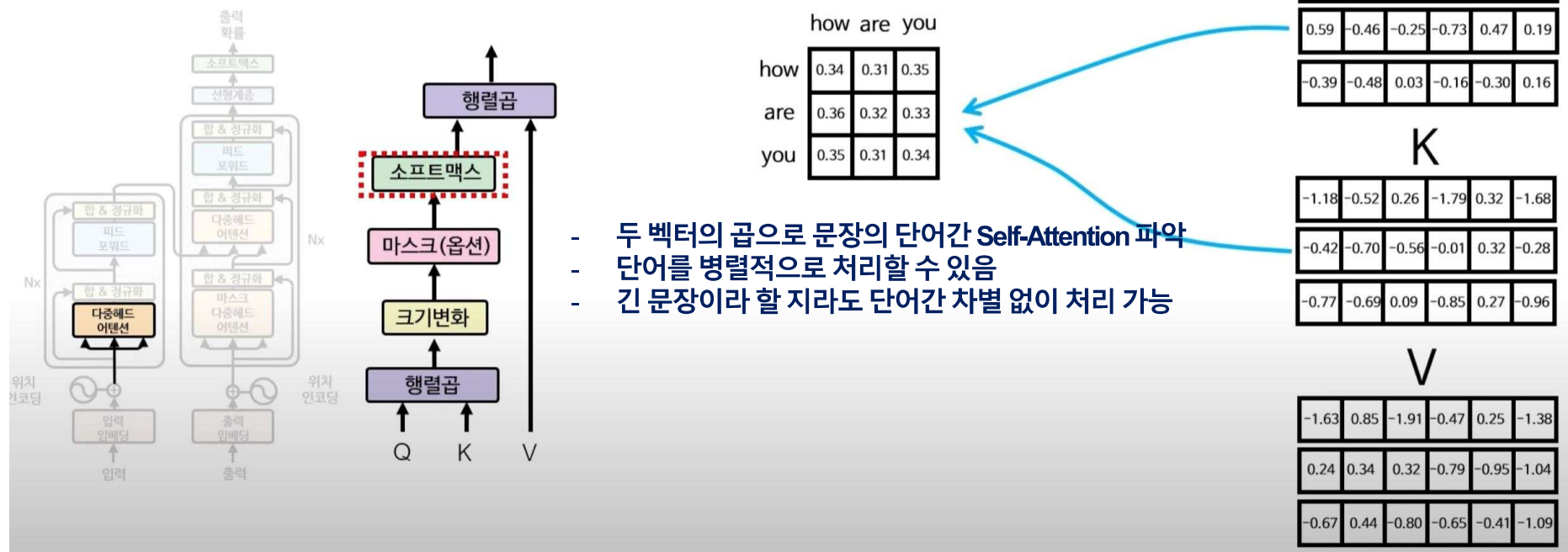
$$\text{softmax}\left(\begin{bmatrix} 0.13 & 0.01 & 0.15 \\ 0.25 & 0.13 & 0.16 \\ 0.26 & 0.14 & 0.22 \end{bmatrix}\right) = \begin{bmatrix} 0.34 & 0.31 & 0.35 \\ 0.36 & 0.32 & 0.33 \\ 0.35 & 0.31 & 0.34 \end{bmatrix}$$



Transformer

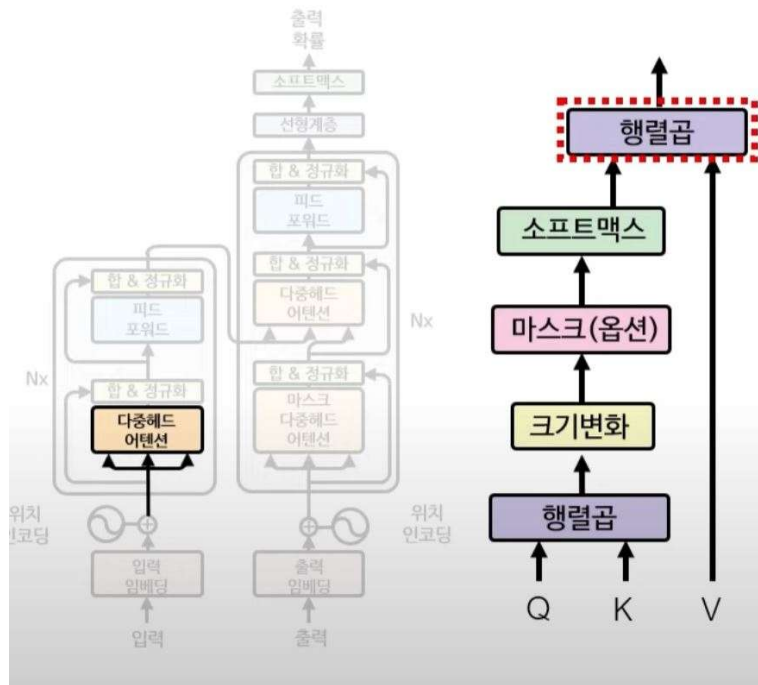


Transformer

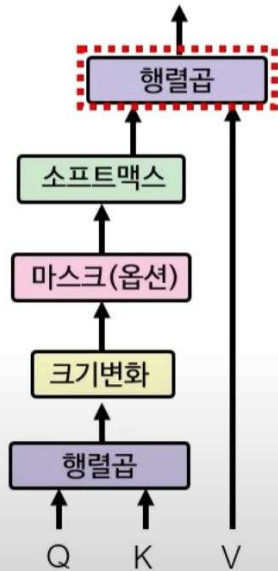


ChatGPT와 gemini 같은 대형 언어모델 LLM을 가능케한 핵심구조

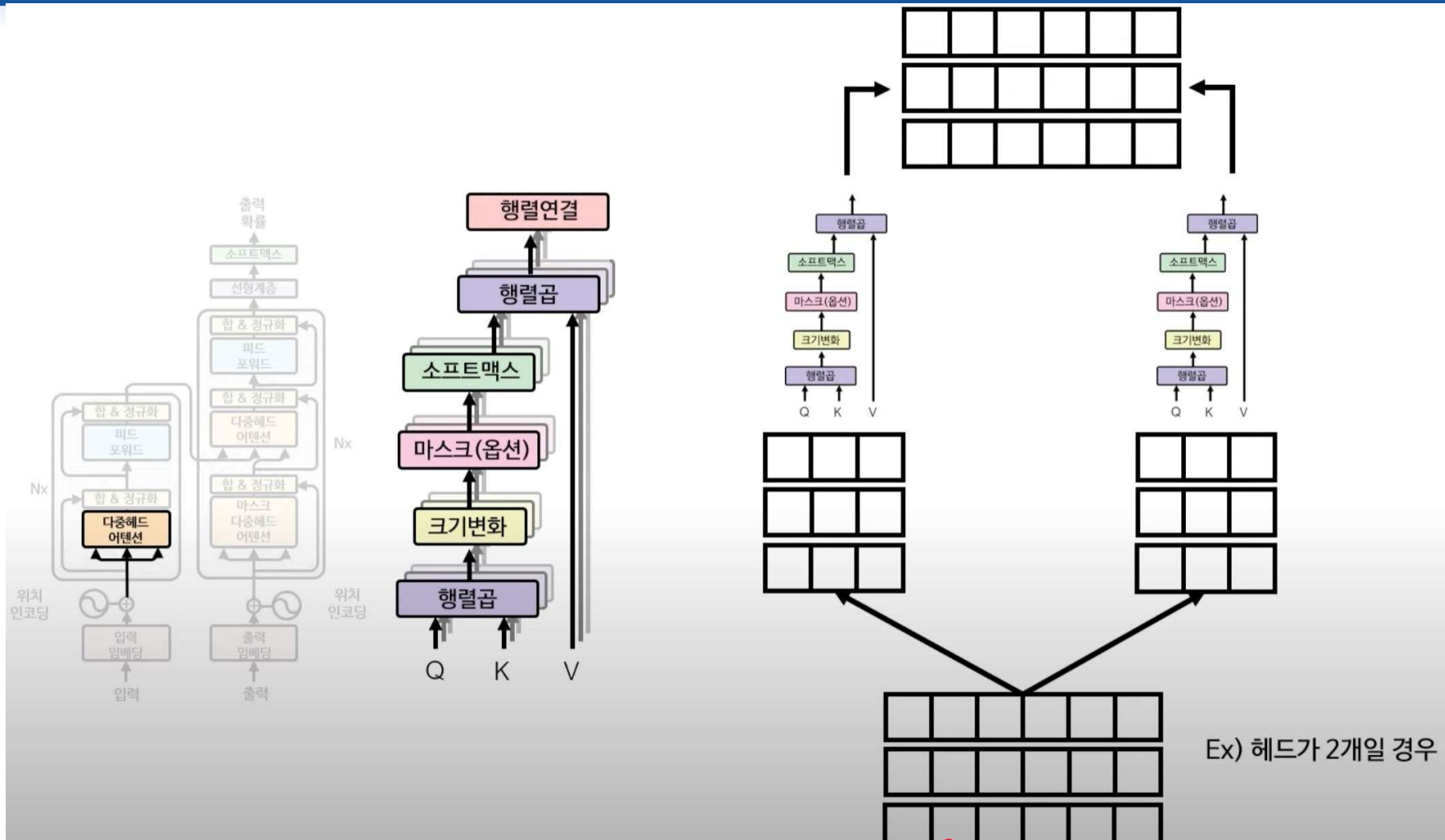
Transformer



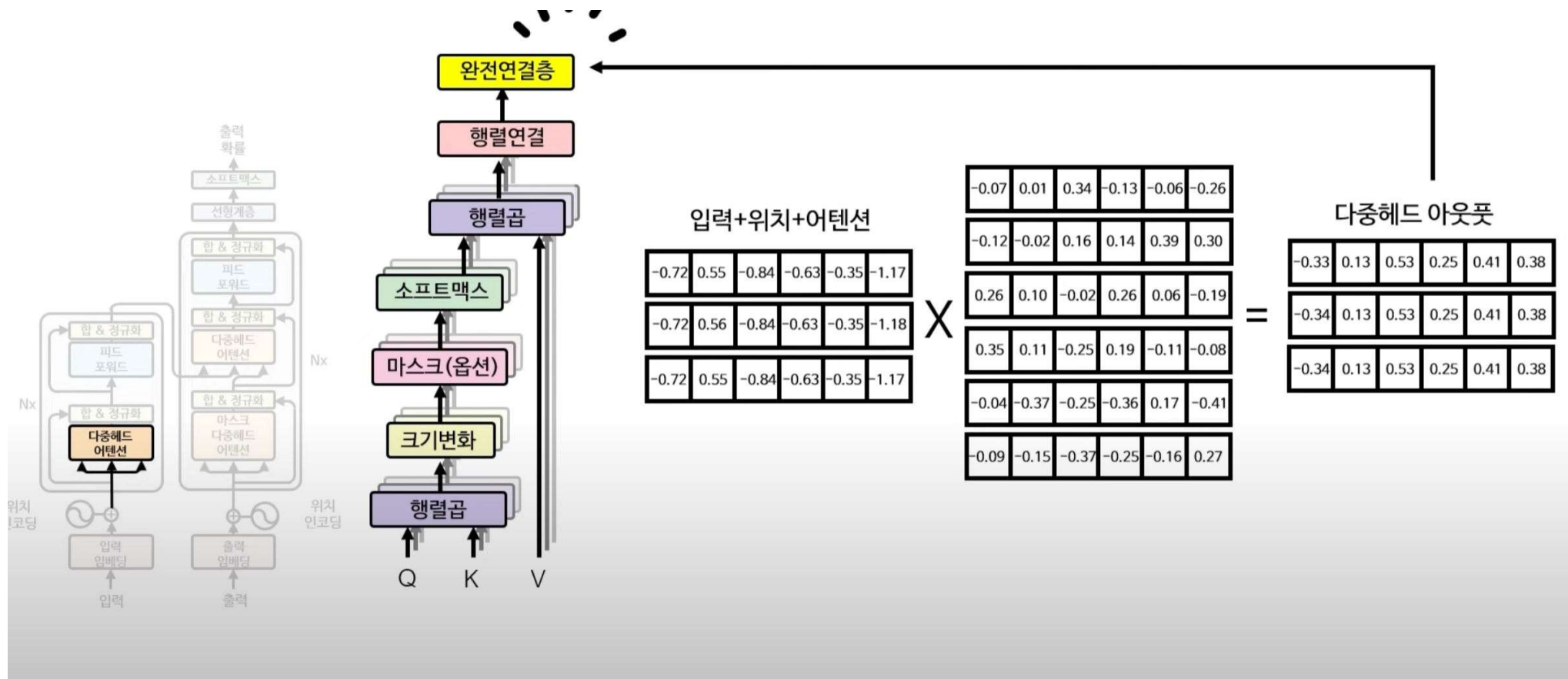
$$\begin{bmatrix} 0.34 & 0.31 & 0.35 \\ 0.36 & 0.32 & 0.33 \\ 0.35 & 0.31 & 0.34 \end{bmatrix} \times \begin{matrix} V \\ \begin{bmatrix} -1.63 & 0.85 & -1.91 & -0.47 & 0.25 & -1.38 \\ 0.24 & 0.34 & 0.32 & -0.79 & -0.95 & -1.04 \\ -0.67 & 0.44 & -0.80 & -0.65 & -0.41 & -1.09 \end{bmatrix} \end{matrix} = \begin{matrix} \text{입력+위치+어텐션} \\ \begin{bmatrix} -0.72 & 0.55 & -0.84 & -0.63 & -0.35 & -1.17 \\ -0.72 & 0.56 & -0.84 & -0.63 & -0.35 & -1.18 \\ -0.72 & 0.55 & -0.84 & -0.63 & -0.35 & -1.17 \end{bmatrix} \end{matrix}$$



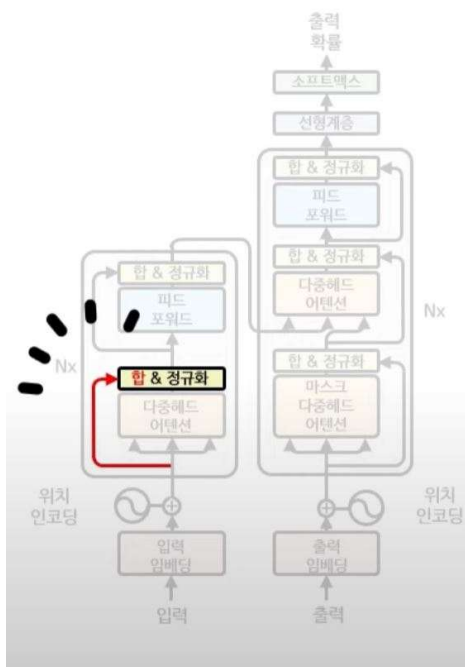
Transformer



Transformer



Transformer



다중헤드 아웃풋

-0.33	0.13	0.53	0.25	0.41	0.38
-0.34	0.13	0.53	0.25	0.41	0.38
-0.34	0.13	0.53	0.25	0.41	0.38

입력+위치 임베딩

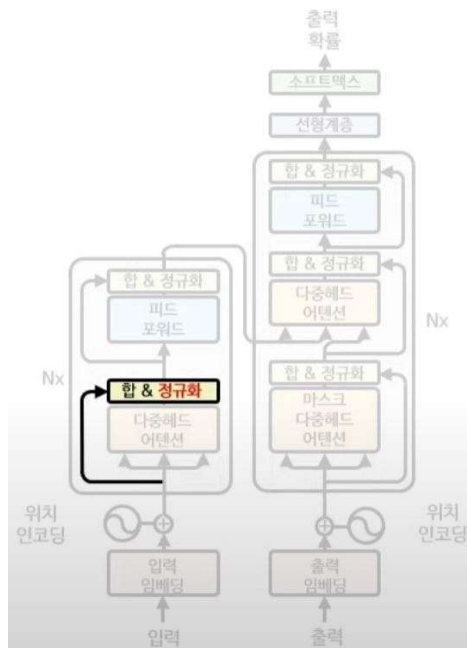
-0.25	1.86	-1.38	0.13	-0.22	2.72
-0.07	-0.12	2.26	1.52	0.35	0.80
-0.15	0.86	0.24	1.23	0.06	1.43

+

=

-0.59	1.99	-0.86	0.38	0.18	3.10
-0.41	0.02	2.79	1.77	0.76	1.19
-0.48	0.99	0.77	1.48	0.47	1.81

Transformer



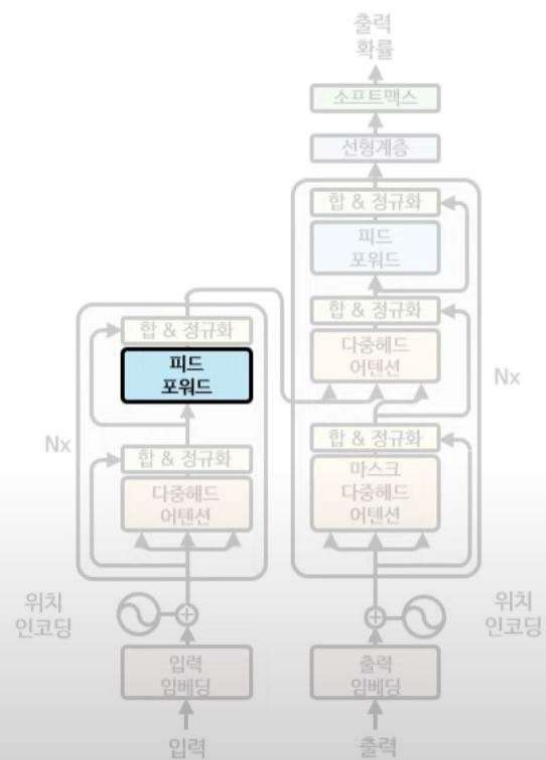
-0.59	1.99	-0.86	0.38	0.18	3.10
-0.41	0.02	2.79	1.77	0.76	1.19
-0.48	0.99	0.77	1.48	0.47	1.81

평균	표준편차
0.70	1.41
1.02	1.07
0.84	0.74

-0.92	0.92	-1.11	-0.23	-0.37	1.70
-1.33	-0.94	1.66	0.70	-0.25	0.16
-1.79	0.21	-0.10	0.86	-0.50	1.32

$$\frac{(\text{표준편차})}{\text{표준편차}}$$

Transformer



-0.92	0.92	-1.11	-0.23	-0.37	1.70
-1.33	-0.94	1.66	0.70	-0.25	0.16
-1.79	0.21	-0.10	0.86	-0.50	1.32

X

0.31	0.20	-0.35	-0.07	-0.13	0.23
0.15	0.34	-0.38	-0.29	-0.30	-0.28
-0.28	0.21	0.17	-0.27	-0.07	0.21
-0.40	0.10	-0.20	0.14	-0.20	0.19
-0.33	-0.00	-0.08	-0.12	-0.12	0.29
0.30	-0.31	-0.23	0.25	-0.39	-0.31

+

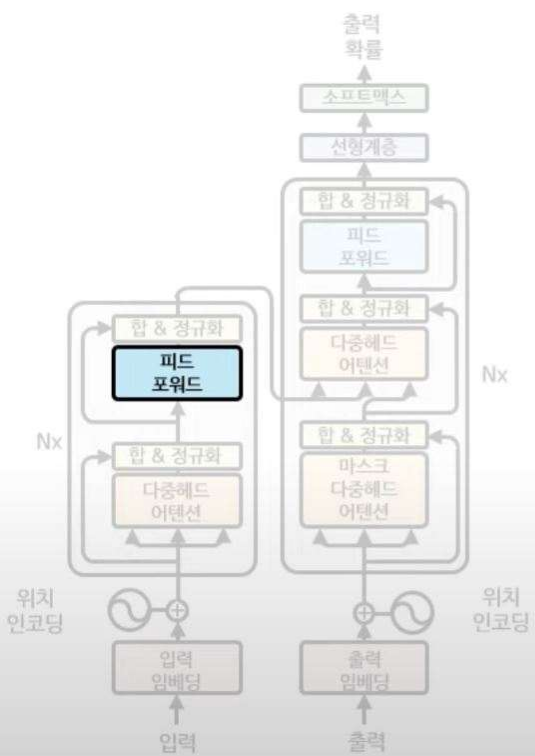
1층 편향

0.29	-0.19	0.15	0.38	-0.06	-0.00
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1.20	-0.85	-0.39	0.93	-0.70	-1.38
-0.89	-0.40	1.10	0.47	0.11	0.31
0.02	-0.81	0.24	0.99	-0.50	-0.89

Transformer



-0.92	0.92	-1.11	-0.23	-0.37	1.70
-1.33	-0.94	1.66	0.70	-0.25	0.16
-1.79	0.21	-0.10	0.86	-0.50	1.32

X

1층 가중치

0.31	0.20	-0.35	-0.07	-0.13	0.23
0.15	0.34	-0.38	-0.29	-0.30	-0.28
-0.28	0.21	0.17	-0.27	-0.07	0.21
-0.40	0.10	-0.20	0.14	-0.20	0.19
-0.33	-0.00	-0.08	-0.12	-0.12	0.29
0.30	-0.31	-0.23	0.25	-0.39	-0.31

+

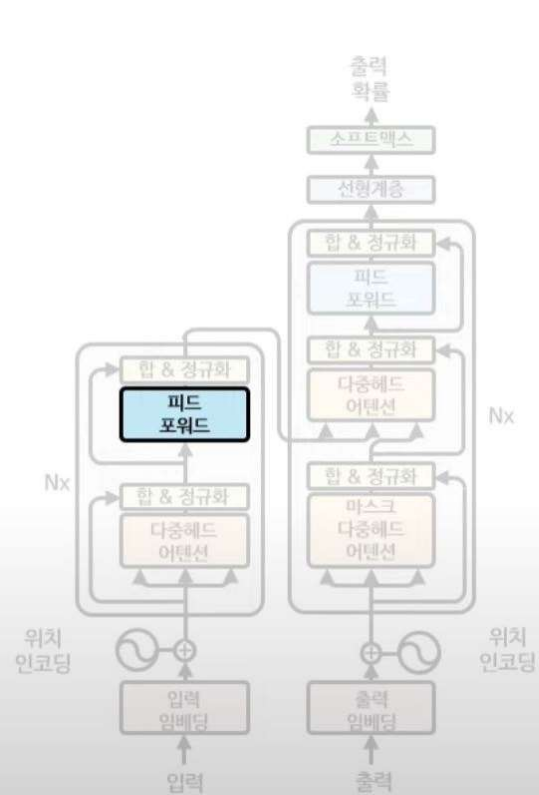
1층 편향

0.29	-0.19	0.15	0.38	-0.06	-0.00
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$$\text{ReLU} \left(\begin{array}{|c|c|c|c|c|c|} \hline 1.20 & -0.85 & -0.39 & 0.93 & -0.70 & -1.38 \\ \hline -0.89 & -0.40 & 1.10 & 0.47 & 0.11 & 0.31 \\ \hline 0.02 & -0.81 & 0.24 & 0.99 & -0.50 & -0.89 \\ \hline \end{array} \right)$$

Transformer



-0.92	0.92	-1.11	-0.23	-0.37	1.70
-1.33	-0.94	1.66	0.70	-0.25	0.16
-1.79	0.21	-0.10	0.86	-0.50	1.32

X

1층 가중치

0.31	0.20	-0.35	-0.07	-0.13	0.23
0.15	0.34	-0.38	-0.29	-0.30	-0.28
-0.28	0.21	0.17	-0.27	-0.07	0.21
-0.40	0.10	-0.20	0.14	-0.20	0.19
-0.33	-0.00	-0.08	-0.12	-0.12	0.29
0.30	-0.31	-0.23	0.25	-0.39	-0.31

+

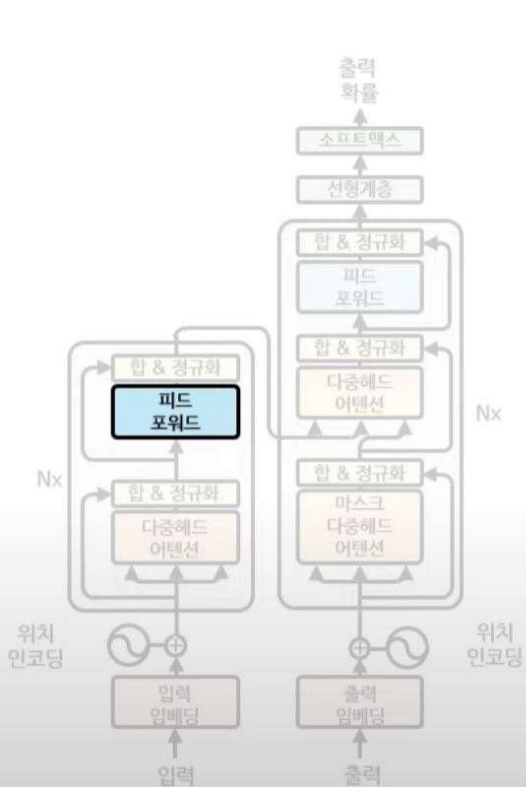
1층 편향

0.29	-0.19	0.15	0.38	-0.06	-0.00
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1.20	0.00	0.00	0.93	0.00	0.00
0.00	0.00	1.10	0.47	0.11	0.31
0.02	0.00	0.24	0.99	0.00	0.00

Transformer



1.20	0.00	0.00	0.93	0.00	0.00
0.00	0.00	1.10	0.47	0.11	0.31
0.02	0.00	0.24	0.99	0.00	0.00

X

2층 가중치

-0.09	0.39	-0.09	-0.39	-0.37	-0.14
-0.34	-0.10	-0.24	0.33	-0.40	-0.13
0.20	-0.34	-0.04	0.34	0.15	0.01
-0.41	0.09	-0.20	-0.06	-0.22	-0.09
0.25	0.23	-0.17	-0.05	-0.26	-0.14
0.31	-0.41	-0.13	-0.17	-0.03	-0.20

+

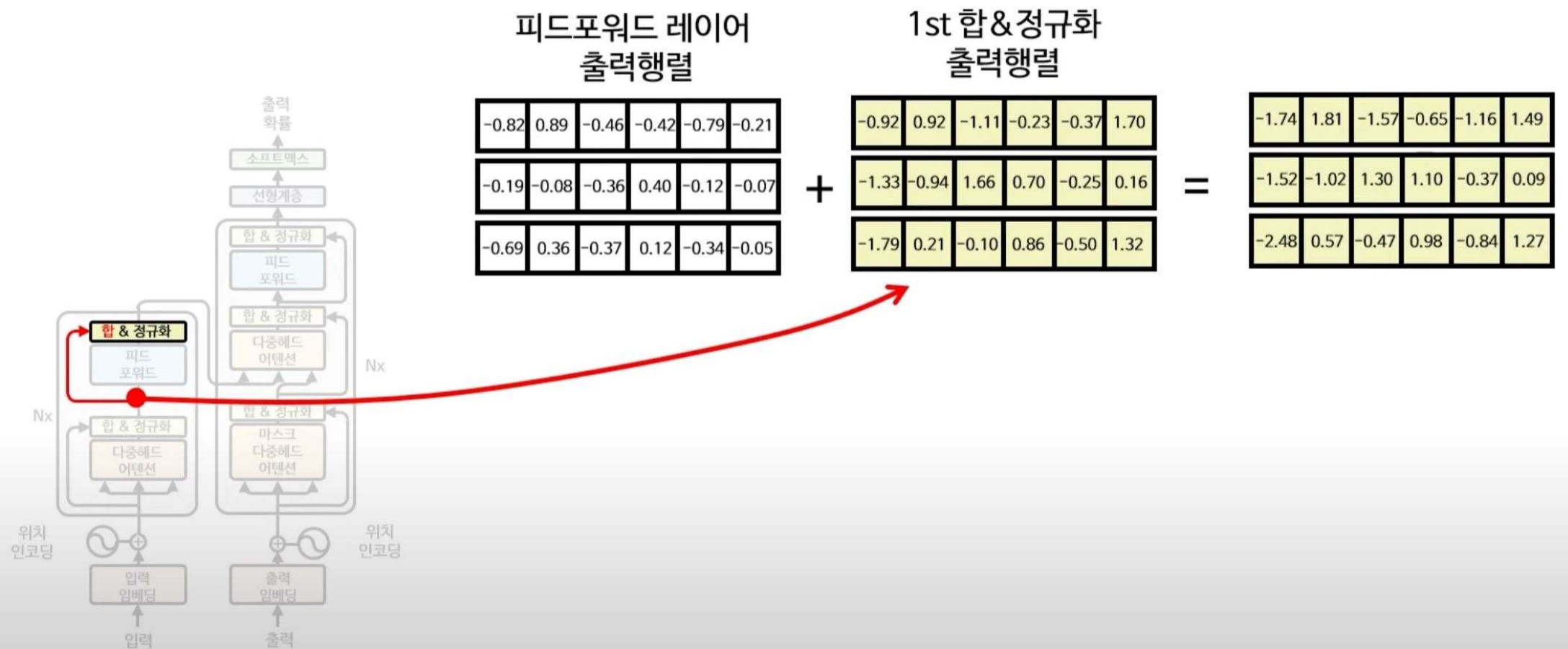
2층 편향

-0.33	0.34	-0.16	0.11	-0.14	0.03
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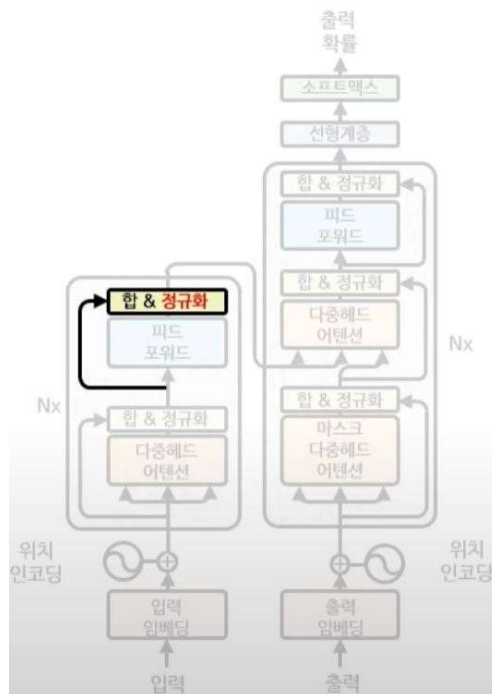


-0.82	0.89	-0.46	-0.42	-0.79	-0.21
-0.19	-0.08	-0.36	0.40	-0.12	-0.07
-0.69	0.36	-0.37	0.12	-0.34	-0.05

Transformer



Transformer



-1.74	1.81	-1.57	-0.65	-1.16	1.49
-1.52	-1.02	1.30	1.10	-0.37	0.09
-2.48	0.57	-0.47	0.98	-0.84	1.27

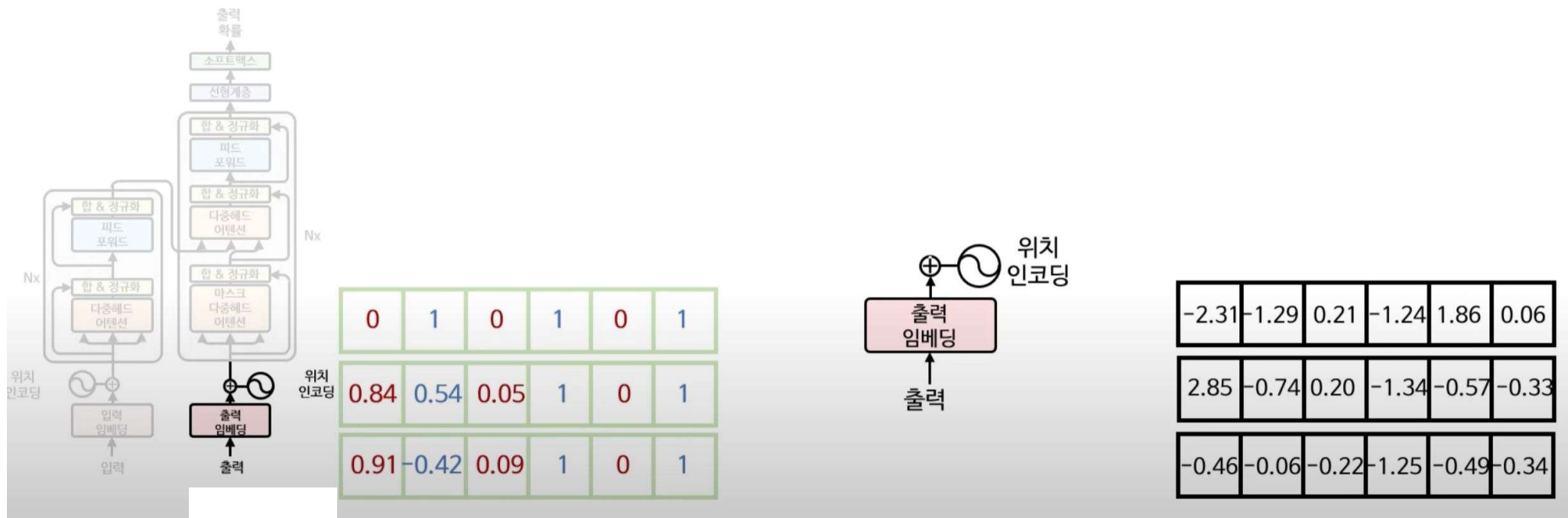
-1.01	1.48	-0.89	-0.24	-0.60	1.26
-1.41	-0.92	1.33	1.13	-0.29	0.15
-1.81	0.57	-0.24	0.89	-0.53	1.12

평균	표준편차
-0.30	1.43
-0.07	1.03
-0.16	1.28

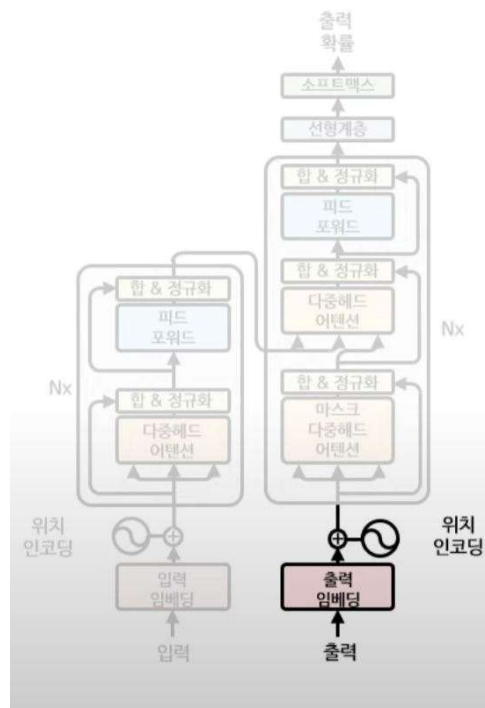
$$\frac{(\text{출력 값들} - \text{평균})}{\text{표준편차}}$$

트랜스포머 인코더의 출력

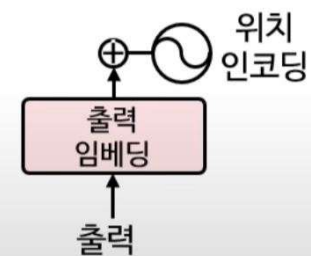
Transformer (디코더)



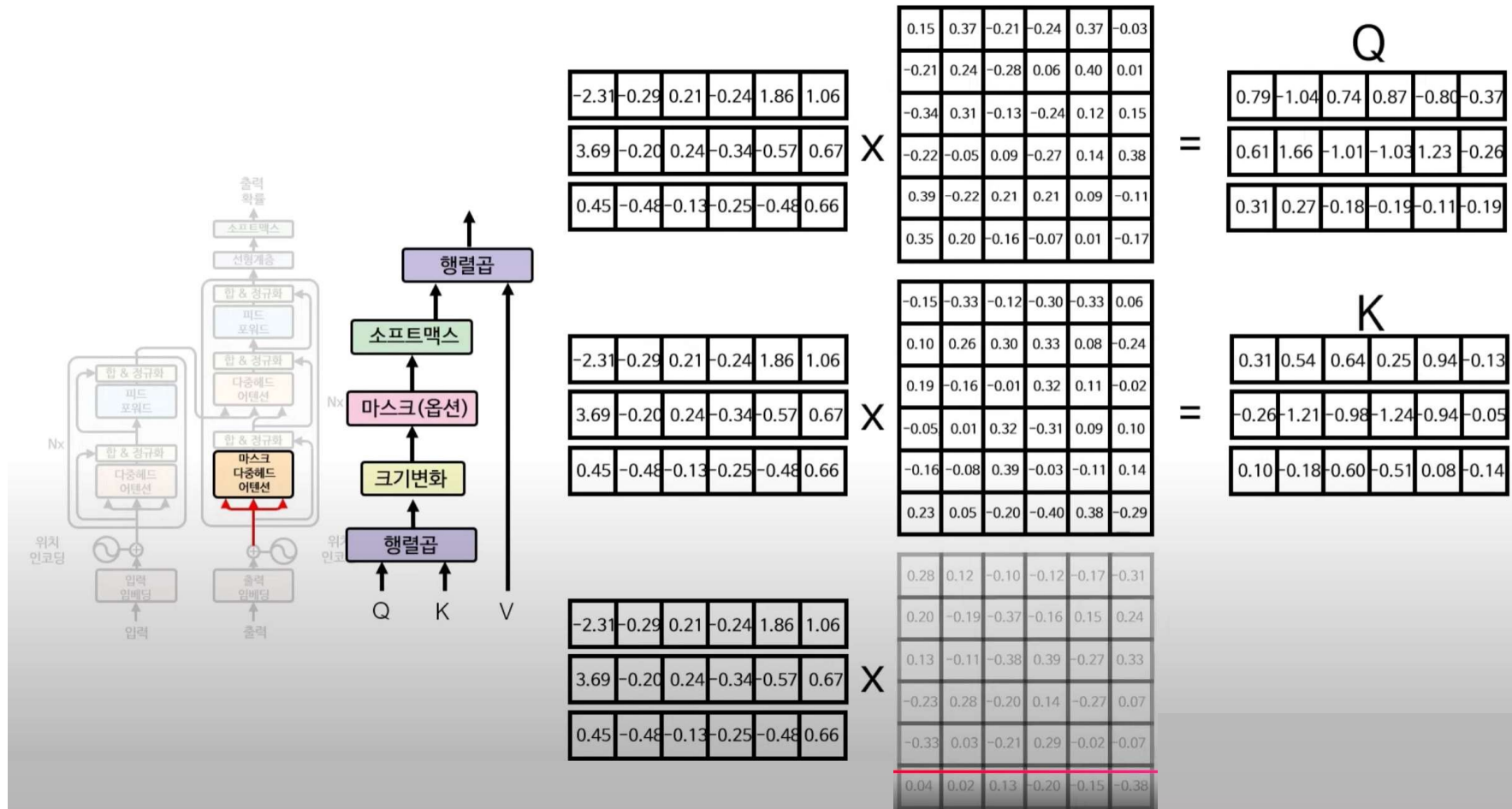
Transformer



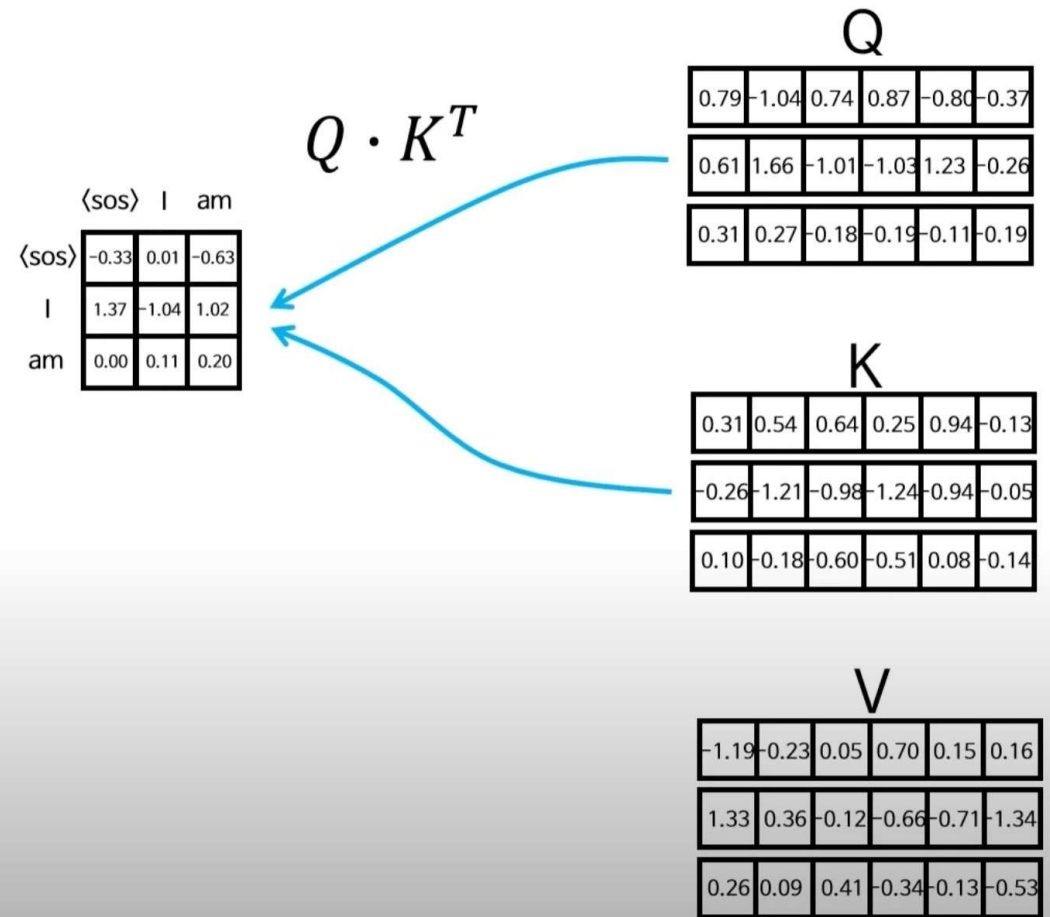
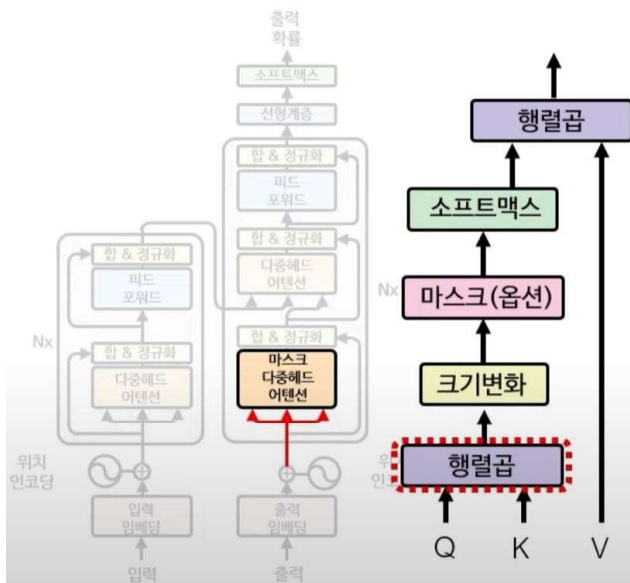
-2.31	-0.29	0.21	-0.24	1.86	1.06
3.69	-0.20	0.24	-0.34	-0.57	0.67
0.45	-0.48	-0.13	-0.25	-0.48	0.66



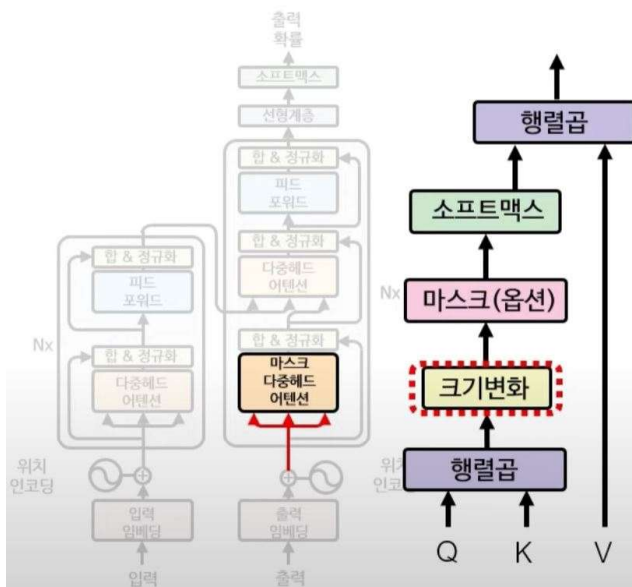
Transformer



Transformer



Transformer



$$\begin{matrix} & \langle \text{sos} \rangle & I & \text{am} \\ \langle \text{sos} \rangle & \begin{bmatrix} -0.33 & 0.01 & -0.63 \\ 1.37 & -1.04 & 1.02 \\ 0.00 & 0.11 & 0.20 \end{bmatrix} \\ I & \\ \text{am} & \end{matrix}$$

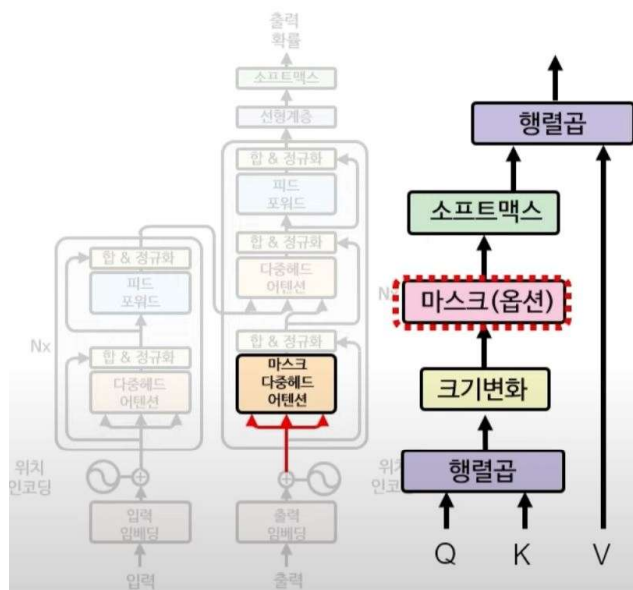
$$\div \sqrt{6} =$$

-0.13	0.00	-0.26
0.56	-0.43	0.42
0.00	0.04	0.08

$$V$$

-1.19	-0.23	0.05	0.70	0.15	0.16
1.33	0.36	-0.12	-0.66	-0.71	-1.34
0.26	0.09	0.41	-0.34	-0.13	-0.53

Transformer

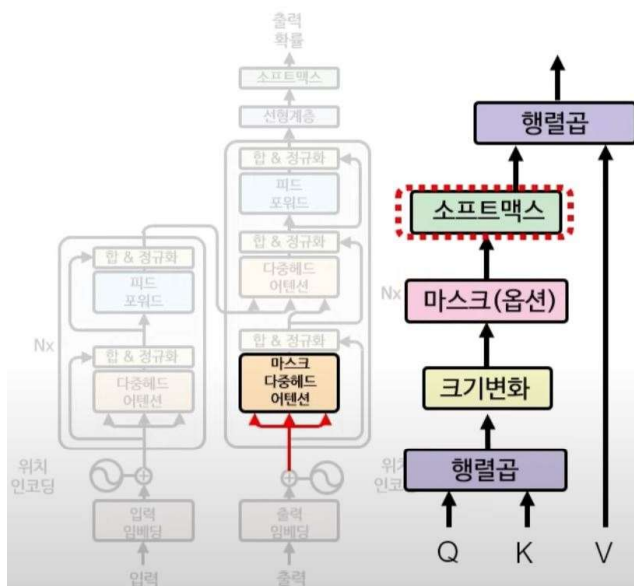


-inf를 넣는 이유는 -inf는 소프트맥스 레이어를 거치면서 0로 바뀌기 때문입니다.

-0.13	-inf	-inf
0.56	-0.43	-inf
0.00	0.04	0.08

V					
-1.19	-0.23	0.05	0.70	0.15	0.16
1.33	0.36	-0.12	-0.66	-0.71	-1.34
0.26	0.09	0.41	-0.34	-0.13	-0.53

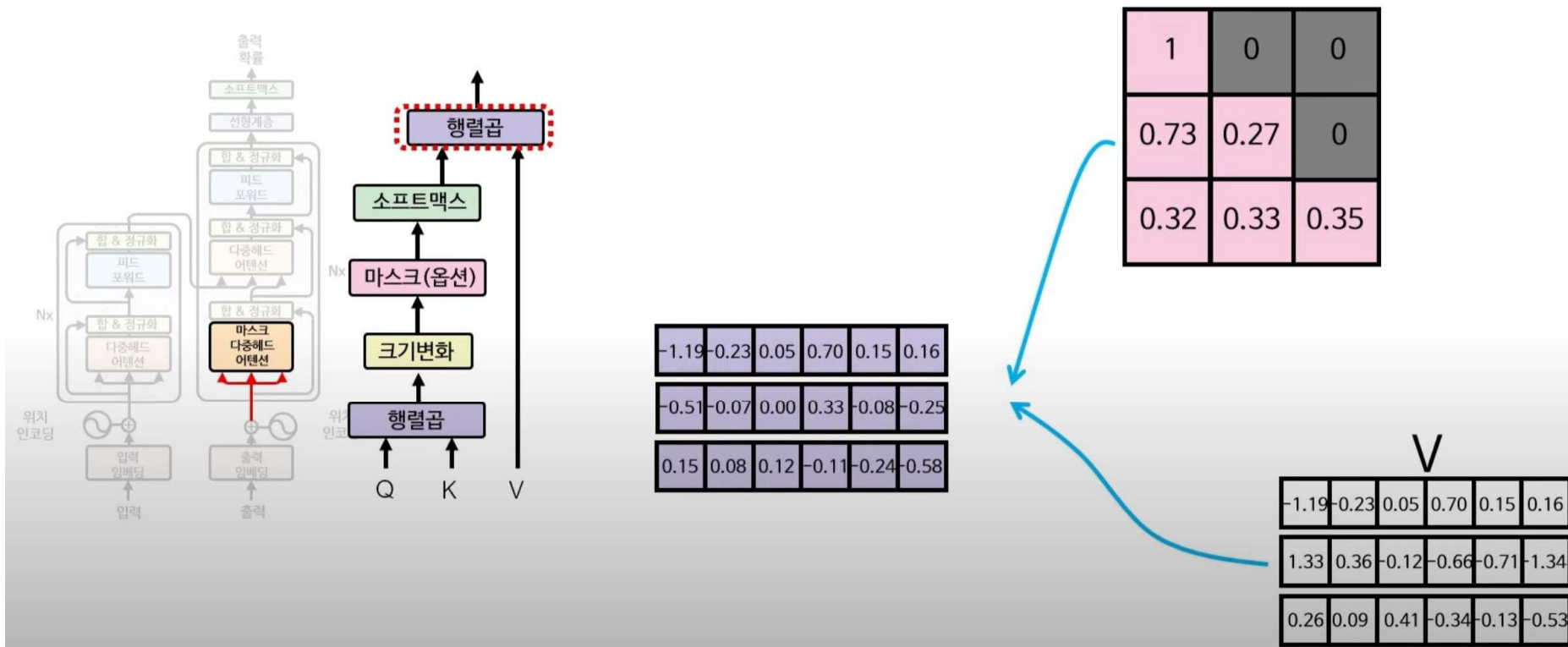
Transformer



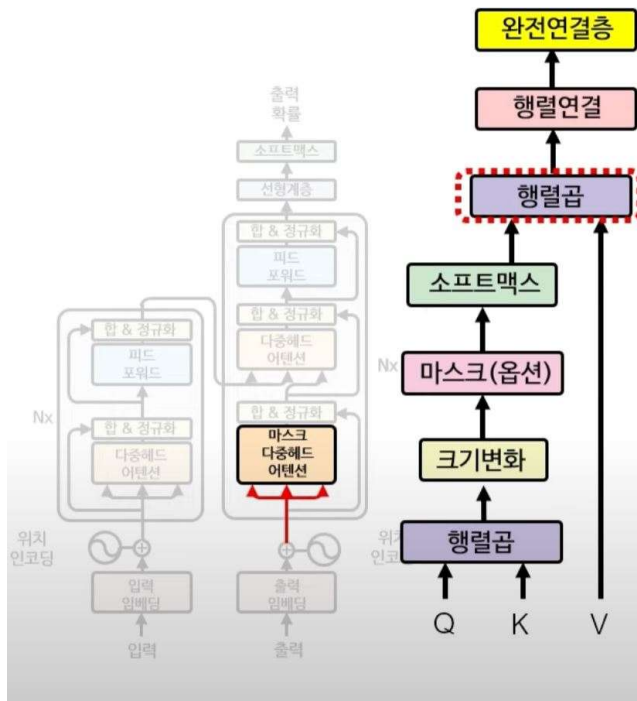
1	0	0
0.73	0.27	0
0.32	0.33	0.35

V					
-1.19	-0.23	0.05	0.70	0.15	0.16
1.33	0.36	-0.12	-0.66	-0.71	-1.34
0.26	0.09	0.41	-0.34	-0.13	-0.53

Transformer

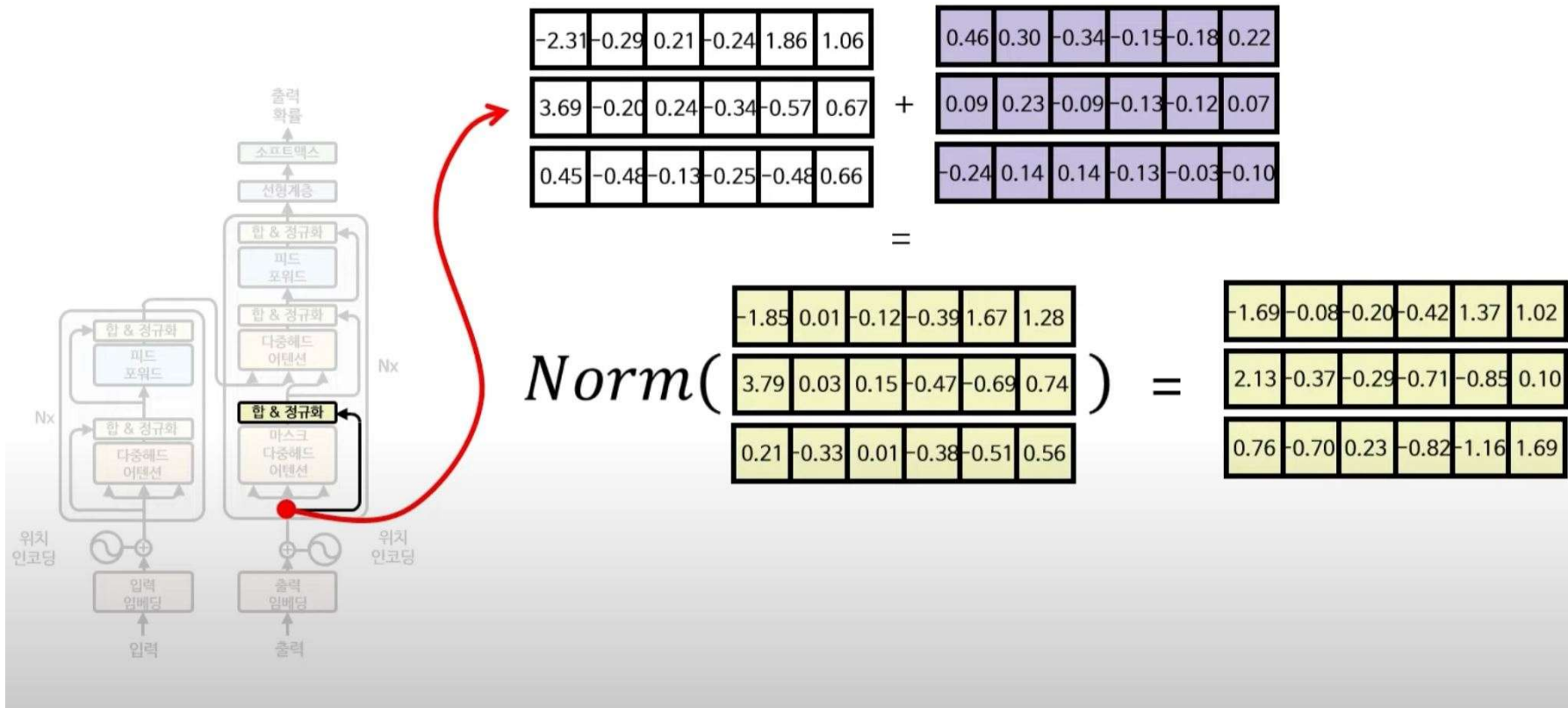


Transformer

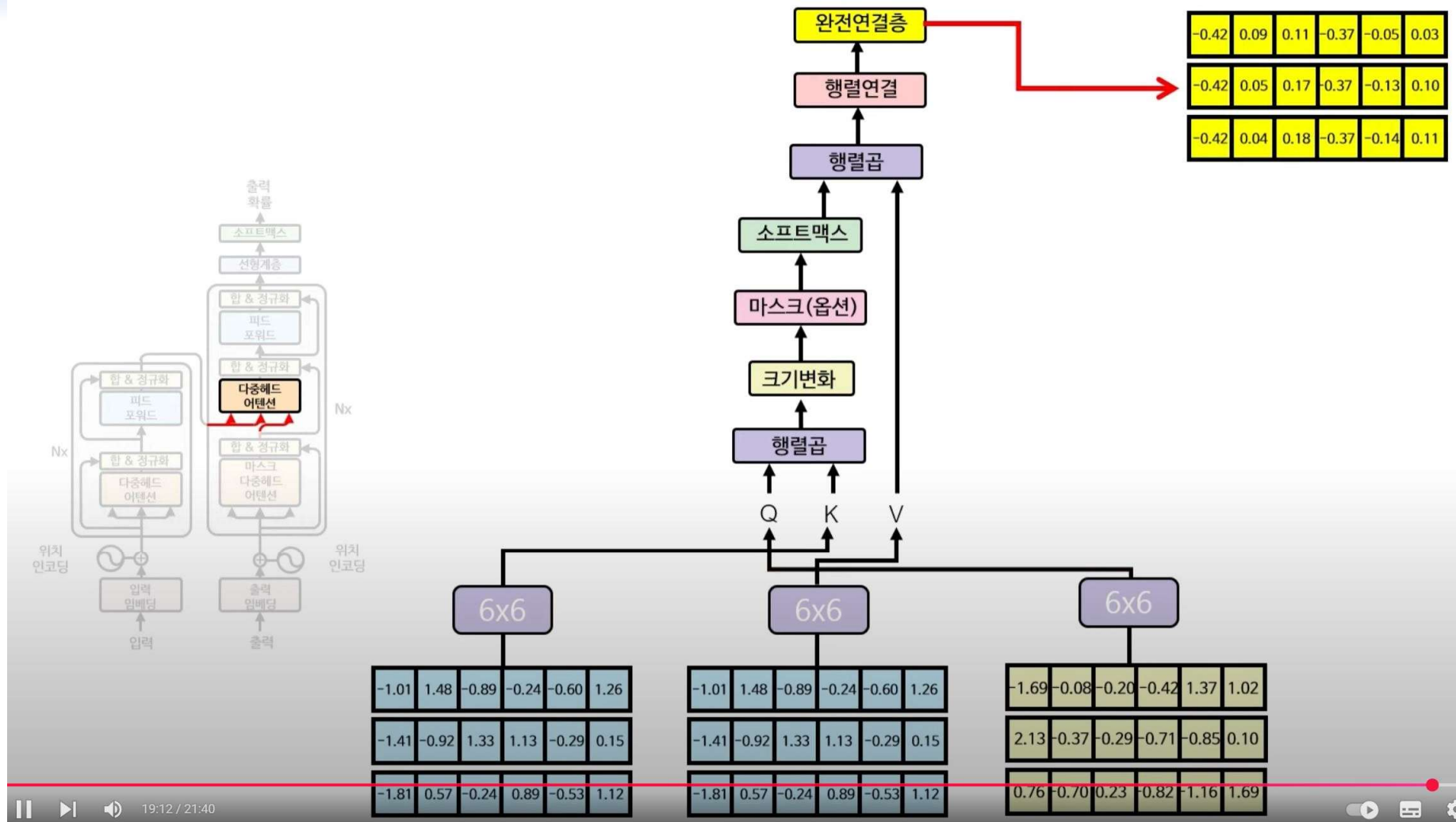


$$\begin{bmatrix} -1.19 & -0.23 & 0.05 & 0.70 & 0.15 & 0.16 \\ -0.51 & -0.07 & 0.00 & 0.33 & -0.08 & -0.25 \\ 0.15 & 0.08 & 0.12 & -0.11 & -0.24 & -0.58 \end{bmatrix} \times \begin{bmatrix} -0.10 & -0.30 & 0.13 & 0.31 & 0.30 & 0.11 \\ -0.20 & -0.22 & -0.31 & -0.01 & -0.35 & -0.20 \\ 0.07 & -0.09 & -0.26 & -0.05 & 0.16 & -0.05 \\ 0.30 & -0.08 & -0.34 & 0.26 & 0.08 & 0.39 \\ 0.32 & 0.03 & 0.36 & -0.04 & 0.09 & 0.27 \\ 0.19 & -0.37 & -0.39 & 0.26 & 0.06 & -0.02 \end{bmatrix} = \begin{bmatrix} 0.46 & 0.30 & -0.34 & -0.15 & -0.18 & 0.22 \\ 0.09 & 0.23 & -0.09 & -0.13 & -0.12 & 0.07 \\ -0.24 & 0.14 & 0.14 & -0.13 & -0.03 & -0.10 \end{bmatrix}$$

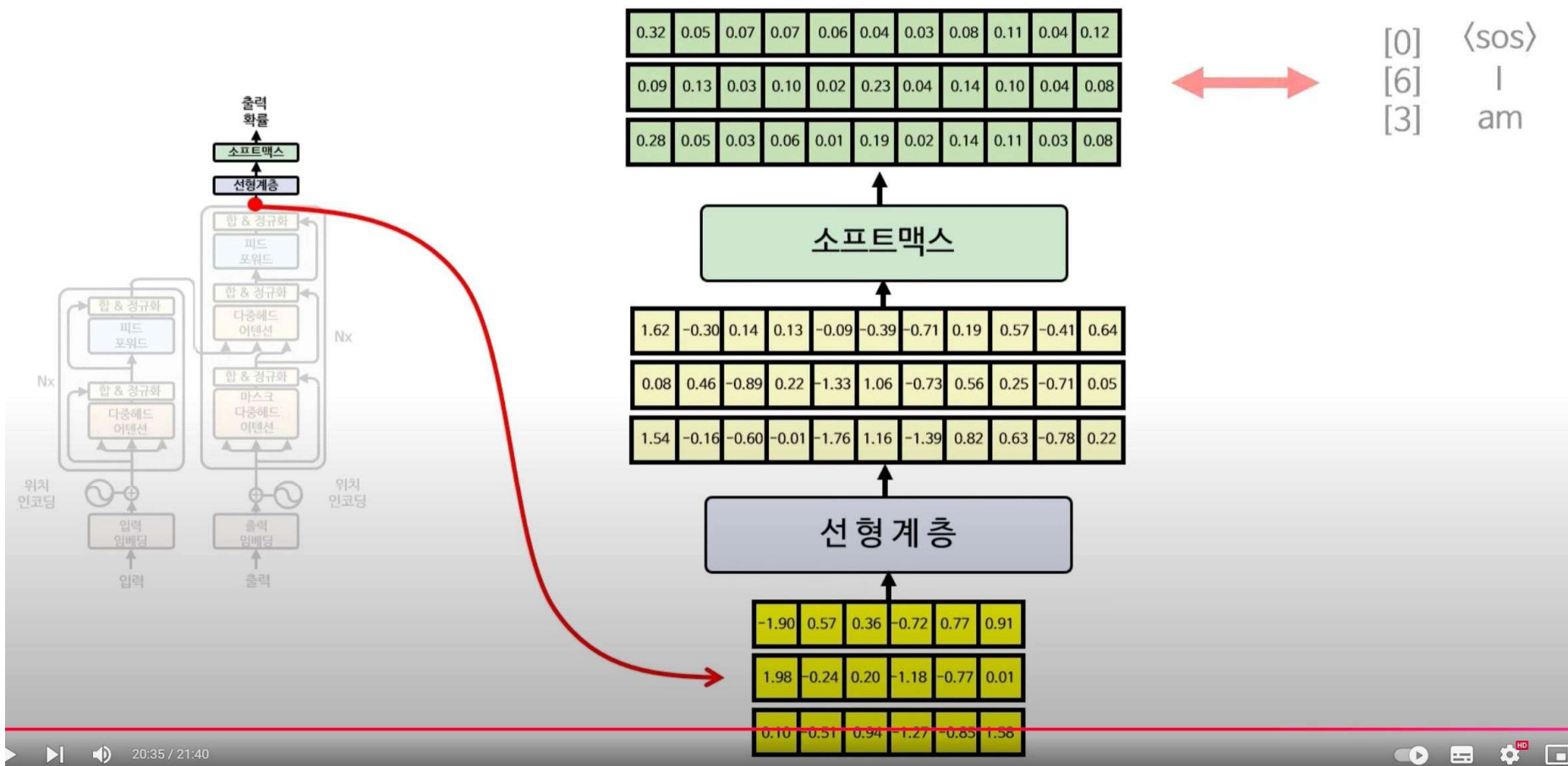
Transformer



Transformer



Transformer



Questions & Answers

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